



Ministry of Health & Family Welfare
Government of India

Annual Report

National Antimicrobial Resistance Surveillance Network (NARS-Net)

Reporting period:
January – December 2025



**National Programme on AMR Containment,
National Centre for Disease Control (NCDC)
Directorate General of Health services
Ministry of Health & Family Welfare
Government of India**

Table of Contents

Acronyms	1
List of Tables	3
List of Figures	4
Executive Summary	6
1. Bacterial AMR Surveillance	8
1.1. Introduction	8
1.2. Findings	11
1.2.1. Data Deduplication	13
1.2.2. Age and Gender distribution of reported AMR data	13
1.2.3. AMR Surveillance of priority pathogens	19
1.2.3.1. Gram-Positive Cocci	19
1.2.3.2. Gram-Negative Bacilli	27
1.2.3.2.1. Enterobacteriaceae	27
1.2.3.2.2. Non-Fermenting Gram-Negative Bacilli	36
1.2.3.2.3. <i>Vibrionaceae</i>	42
1.2.4. AMR Trends	42
1.2.5. Molecular Detection and Characterization of Priority Bacterial Pathogens	47
1.2.6. Discussion	50
2. Fungal AMR Surveillance	53
2.1. Introduction	53
2.2.2 Methodology	53
2.2.3 Details of 997 yeasts characterized at NRL during 2025	54
Annexure I	62

Acronyms

AMR	Antimicrobial Resistance
AFST	Antifungal susceptibility testing
Amox-Clav	Amoxicillin-Clavulanic acid
AST	Antimicrobial Susceptibility Testing
BMD	Broth Microdilution
CBDDR	Centre for Bacterial Diseases and Drug Resistance
CI	Confidence Interval
CLSI	Clinical and Laboratory Standards Institute
CRE	Carbapenem Resistant Enterobacterales
CSV	Comma Separated Value
DMSO	Dimethyl sulfoxide
DNA	Deoxyribonucleic Acid
E. coli	<i>Escherichia coli</i>
EME	Emergency
EQAS	External Quality Assessment Scheme
ESBL	Extended Spectrum β -lactamase
EUCAST	European Committee for Antimicrobial Susceptibility Testing
GBD	Global Burden of Disease
GLASS	Global Antimicrobial Resistance and Use Surveillance System
GNB	Gram-Negative bacilli
GPC	Gram-Positive cocci
ICU	Intensive Care Unit
IPD	Inpatient Department
IN	Inpatient
ID	Identification
IQC	Internal Quality Control
ITS	Internal Transcribed Spacer
LIMS	Laboratory Information Management System
MALDI-TOF	Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry
MIC	Minimum Inhibitory Concentration

MICU	Medical Intensive Care Unit
MOPS	3-(N-morpholino) propane sulfonic acid
MoHFW	Ministry of Health and Family Welfare
MRSA	Methicillin-Resistant <i>Staphylococcus aureus</i>
NARS-Net	National Antimicrobial Resistance Surveillance Network
NCC	National Coordinating Centre
NCDC	National Centre for Disease Control
NFGNB	Non-fermenting Gram-negative bacilli
NICU	Neonatal Intensive Care Unit
NRL	National Reference Laboratory
OPD	Outpatient Department
OSBF	Other Sterile Body Fluids
OUT	Outpatient
PA	Pus Aspirate
PICU	Pediatric Intensive Care Unit
Pip-Tazo	Piperacillin-Tazobactam
%R	Percentage Resistant
R I S	Resistant Intermediate Sensitive
RPMI	Roswell Park Memorial Institute medium
SICU	Surgical Intensive Care Unit;
SOP	Standard Operating Procedure
TICU	Trauma Intensive Care Unit
TMP-SMX	Trimethoprim-Sulfamethoxazole
UNK	Unknown
UT	Union Territory
VBA	Visual Basic Application
VISA	Vancomycin Intermediate resistant <i>S. aureus</i>
VRE	Vancomycin-Resistant <i>Enterococcus</i> species
VRSA	Vancomycin Resistant <i>S. aureus</i>
WHO	World Health Organization

List of Tables

Table 1: Priority Pathogens and specimens included under Bacterial AMR Surveillance	9
Table 2: Distribution of priority pathogen isolates based on specimen type (N=2,17,513).....	14
Table 3: Specimen type wise distribution of isolates (N= 2,17,513)	15
Table 4: Proportion of priority pathogens isolated from various specimen types (N= 2,17,513)	16
Table 5: Distribution of priority pathogen isolates by location type (N=217513)	17
Table 6: Resistance profile of Staphylococcus aureus (N=23,507)	20
Table 7: Resistance profile of Enterococcus species (N=22,318)	24
Table 8: Resistance profile of Escherichia coli (N=68,341)	28
Table 9: Resistance profile of Klebsiella species (N=52,597)	32
Table 10: Resistance profile of Salmonella enterica Typhi and Paratyphi (N=1145) from blood	34
Table 11: Resistance profile of Shigella species (N=120) from stool specimens	35
Table 12: Resistance profile of Pseudomonas species (N=26,637)	37
Table 13: Resistance profile of Acinetobacter species (N=22,711)	40
Table 14: Resistance profile of Vibrio cholerae (N=106)	42
Table 15: Distribution of other (8%) yeast species characterized at NRL	55
Table 16: Distribution of different yeast species by location type in ICU and emergency	57
Table 17: Distribution of different yeast species by location type in Inpatient Wards and Outpatient	57
Table 18: Distribution of different Candida species in different types of Intensive Care Units (ICUs)	58

List of Figures

Figure 1: Geographical Location of National AMR Surveillance Network laboratories which have submitted AMR data from January to December 2025	10
Figure 2: Flowchart to Obtain a Cleaned WHONET Data File for Analysis	12
Figure 3: Distribution of unique patients according to gender (N=2,02,573)	13
Figure 4: Distribution of priority pathogen isolates by age category (N=2,17,513)	14
Figure 5: Distribution of isolates by location type (N=217513)	16
Figure 6: Distribution of priority pathogen isolates by location-type.....	18
Figure 7: Resistance profile of <i>S. aureus</i> in blood (N=6775) by location type	21
Figure 8: Resistance profile of Methicillin Resistant <i>S. aureus</i> (MRSA) in blood (N=3,251).....	22
Figure 9: Resistance profile of <i>Enterococcus</i> species in blood (N=3,677) by location type	25
Figure 10: Resistance profile of <i>Enterococcus</i> species in urine (N=15,266) by location type	26
Figure 11: Resistance profile of <i>Escherichia coli</i> isolated from blood (N=3,795) by location type	29
Figure 12: Resistance profile of <i>Escherichia coli</i> isolated from urine (N=47,226) by location type	30
Figure 13: Resistance profile of <i>Klebsiella</i> species isolated from blood (N=9,478) by location type....	33
Figure 14: Resistance profile of <i>Pseudomonas</i> species isolated from blood (N=4,478) by location type	38
Figure 15: Resistance profile of <i>Acinetobacter</i> species isolated from blood (N=8,890) by location type	41
Figure 16: Trends of Methicillin-resistant <i>S. aureus</i> (MRSA) isolated from blood (2017-2025)	43
Figure 17: Trends of Linezolid resistant <i>Staphylococcus aureus</i> isolated from blood (2018 to 2025) .	43
Figure 18: Trends of Vancomycin resistant <i>Enterococcus</i> sp. (VRE) isolated from blood (2018-2025)	44
Figure 19: Trends of Extended spectrum beta-lactamase (ESBL) producing <i>E. coli</i> isolated from blood (2017-2025)	44
Figure 20: Trends of Extended-Spectrum Beta-Lactamase (ESBL) producing <i>Klebsiella</i> spp. in blood (2017-2025)	45
Figure 21: Four-year trend analysis (2022-2025) of Carbapenemase producing <i>E. coli</i> and <i>Klebsiella</i> species isolated from blood.....	45

Figure 22: Four-year trend analysis (2022-2025) of Carbapenemase producing <i>Pseudomonas</i> species and <i>Acinetobacter</i> species isolated from blood	46
Figure 23: Distribution of NDA & OXA-48 gene in <i>Klebsiella</i> species	48
Figure 24: Distribution of NDA & OXA-48 gene in <i>E. coli</i>	48
Figure 25: Distribution of NDA & OXA-48 gene in <i>Acinetobacter</i> species	49
Figure 26: Distribution of NDA & OXA-48 gene in <i>Pseudomonas</i> species	49
Figure 27: Distribution of vanA & vanB genes in <i>Enterococcus</i> species	50
Figure 28: The pie chart illustrating the percentage distribution of various yeast species identified by MALDI-TOF MS and ITS sequencing at NRL, VPCI	54
Figure 29: Gender-wise distribution of patients from whom yeast isolates (n=997) were obtained ..	55
Figure 30: Age distribution of patients from whom yeast isolates were recovered	56
Figure 31: Bar graph representing resistance percentages of <i>Candida</i> spp. against to fluconazole ...	60

Executive Summary

Antimicrobial Resistance (AMR) surveillance is crucial for tracking and combating the growing threat of AMR. This is the ninth report of National AMR Surveillance Network (NARS-Net) established under the National Programme on AMR Containment coordinated by the National Centre for Disease Control (NCDC), MoHFW, India. This report includes bacterial AMR surveillance data from 62 sentinel surveillance sites. National programme on AMR containment has been expanded in a phased manner and currently includes 70 laboratories in 34 states/UTs (as of March 2026). This report provides insights into antimicrobial resistance patterns, trends and emerging resistance across the network sites.

The current report includes data of bacterial and fungal AMR Surveillance for the reporting period from 01st January 2025 to 31st December 2025. Standard operating procedures (SOPs) on antimicrobial susceptibility testing (AST), internal quality control (IQC), AMR data management developed under the programme are implemented at NARS-Net sites to ensure quality data. The panel of antimicrobials used for surveillance is standardized in alignment with the international standards and prescription practices of clinicians in the country. Trainings and workshops are conducted regularly for NARS-Net sites to ensure compliance to the SoPs under the programme. Wet-lab trainings on colistin agar dilution, vancomycin screen agar and colistin and vancomycin broth microdilution (BMD) testing, both at the NRL, NCDC and onsite during site visits, have contributed to an increase in the number of sentinel sites performing MIC determination and submitting quality data. Online trainings are also organized on regular basis to reinforce the testing standards at laboratories. Monthly feedbacks through data quality monitoring calls has played a crucial role in improving compliance to data standards defined under the programme. Enrolment of sites under External Quality Assessment Scheme (EQAS) and bacterial identification and AST confirmations of a defined number of isolates conducted by AMR-NRLs, has played a significant role in improving data quality.

The key findings of this report are:

- Over the years, with increasing number of sites, there has been a gradual increase in the number of reported isolates, rising from 25,833 in 2017 to 2,17,513 in the current data reporting period.
- The most commonly isolated priority bacterial pathogen was *E. coli* (31%), which is similar to the previous five years, followed by *Klebsiella* species (24%), *Pseudomonas* species (12%),

S. aureus (11%), *Acinetobacter* species (10%) *Enterococcus* species (10%) and *Salmonella enterica* serovar Typhi and Paratyphi (0.5%).

- Among the specimen types, urine (44%) was the most frequently reported specimen followed by pus aspirate (31%) and blood (18%). From urine specimens, *E. coli* (49%) was the most frequently isolated pathogen, while *Klebsiella* species was the most commonly isolated pathogen from blood specimens (25%) and Other sterile body fluids (26%) while from pus specimens *Klebsiella* species (24%) and *S. aureus* (23%) both were the predominant isolates.
- Majority of isolates of *Enterococcus* species were from urine specimens (68%) and most *S. aureus* isolates were from pus aspirates (67%) consistent with the previous reports.
- Proportion of MRSA in blood was slightly higher from last year in all settings, from 63% to 65% in intensive care units, from 55% to 61% among inpatients and from 44% to 51% in outpatients.
- Proportion of VRE among blood isolates was found to be similar to that during last year, it was 20% in ICU setting and 19% among inpatients.
- Among *E. coli* from blood, 80% of isolates were ESBL producers, and among *Klebsiella* spp. blood isolates, 83% were ESBL producers.
- Also, trend analysis for carbapenem resistance in gram negative bacterial isolates from blood cultures showed a decreasing trend from previous year in *E. coli* (from 43% to 41%), *Pseudomonas* species (35% to 30%) and *Acinetobacter* species (70% to 68%) while carbapenem resistant *Klebsiella* spp. has slightly increased from 59% to 62%.

This report emphasizes on the need for continuous strengthening of surveillance system, targeted interventions, development of evidence-based policies, sustained collaborations and investments to combat AMR. The AMR surveillance data in this report serves as a crucial resource for policy makers, healthcare professionals, and researchers working to address the growing threat of AMR.

The AMR surveillance data in the current report like previous years continues to have the limitation of non-availability of results of complete panel of antibiotics for a given isolate as per the Standard Operating Procedure for Antimicrobial resistance Surveillance. Another important limitation is that the referral of clinical specimens for culture and sensitivity testing is mostly from patients not responding to the prescribed antibiotics. These limitations have the potential to skew the AMR trends to higher side than the actual trends.

1. Bacterial AMR Surveillance

1.1. Introduction

Antimicrobial resistance (AMR) is a rapidly escalating global health threat. As pathogens evolve to resist the existing antimicrobials, once-manageable diseases become increasingly difficult or impossible to cure. Global Burden of Disease 2021 report by Antimicrobial Resistance Collaborators published in *The Lancet* (2024) estimated that 4.71 million deaths were associated with bacterial AMR including 1.14 million deaths attributable to bacterial AMR.¹ The report also states that among Gram-negative bacteria, resistance to carbapenems increased more than any other antibiotic class, rising from 0.619 million deaths in 1990 to 1.03 million associated deaths in 2021. The report also forecasted that an estimated 1.91 million deaths attributable to AMR and 8.22 million deaths associated with AMR could occur globally in 2050. These figures highlight the immense burden of AMR on public health, particularly in low- and middle-income countries where access to quality healthcare and diagnostics is limited. Beyond mortality, AMR also leads to prolonged illness, increased healthcare costs, and greater risk of complications, posing a serious threat to modern medicine, including surgery, cancer therapy and intensive care. Without urgent and coordinated action, AMR will continue to undermine modern medicine and threaten global health security. To monitor AMR, a robust surveillance system is needed in the global fight against drug-resistant infections, providing the critical data needed to track resistance patterns, update treatment guidelines, and guide public health interventions.

To address the problem of AMR, Government of India initiated the National Programme on AMR containment in 2013, during the 12th (2012-2017) five-year plan which is being coordinated by the National Centre for Disease Control (NCDC). Under the programme, the National AMR Surveillance Network (NARS-Net) has been established to monitor real time AMR trends and to generate annual reports on National AMR Surveillance data obtained from NARS-Net sites. The current report is the ninth annual report of AMR Surveillance data which includes data from 62 network sites (Fig. 1).

NARS-Net has been expanded in a phased manner to have representation from all states and Union Territories in the country. Currently, NARS-Net includes 70 medical college laboratories from 28 states and 6 UTs (as of March 2026) with adequate geographical representation. The NARS-Net sentinel surveillance sites conduct laboratory-based AMR surveillance of nine priority bacterial pathogens namely *Staphylococcus aureus*, *Enterococcus* species, *Escherichia coli*, *Klebsiella* species, *Pseudomonas* species, *Acinetobacter* species, *Salmonella enterica* serotype Typhi and Paratyphi, *Shigella* species and *Vibrio cholerae* (Table 1).

¹ Naghavi, Mohsen et al. Global burden of bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050. *The Lancet*, Volume 404, Issue 10459, 1199 - 1226

The AMR surveillance data of these priority pathogens submitted by the NARS-Net sites to AMR programme unit at NCDC is being collated and analyzed at NCDC after validation using WHONET, a microbiology data management open-source offline software. After data validation, the semi-annual bulletins² and the annual National AMR surveillance reports³ are developed which are available on NCDC website for use by various stakeholders at national and state level. The network data is also submitted annually to the World Health Organization's Global AMR Surveillance and Use System (WHO-GLASS) since 2018 by NCDC which is designated by MoHFW as the National Coordinating Centre for AMR Surveillance in India.

Table 1: Priority Pathogens and specimens included under Bacterial AMR Surveillance

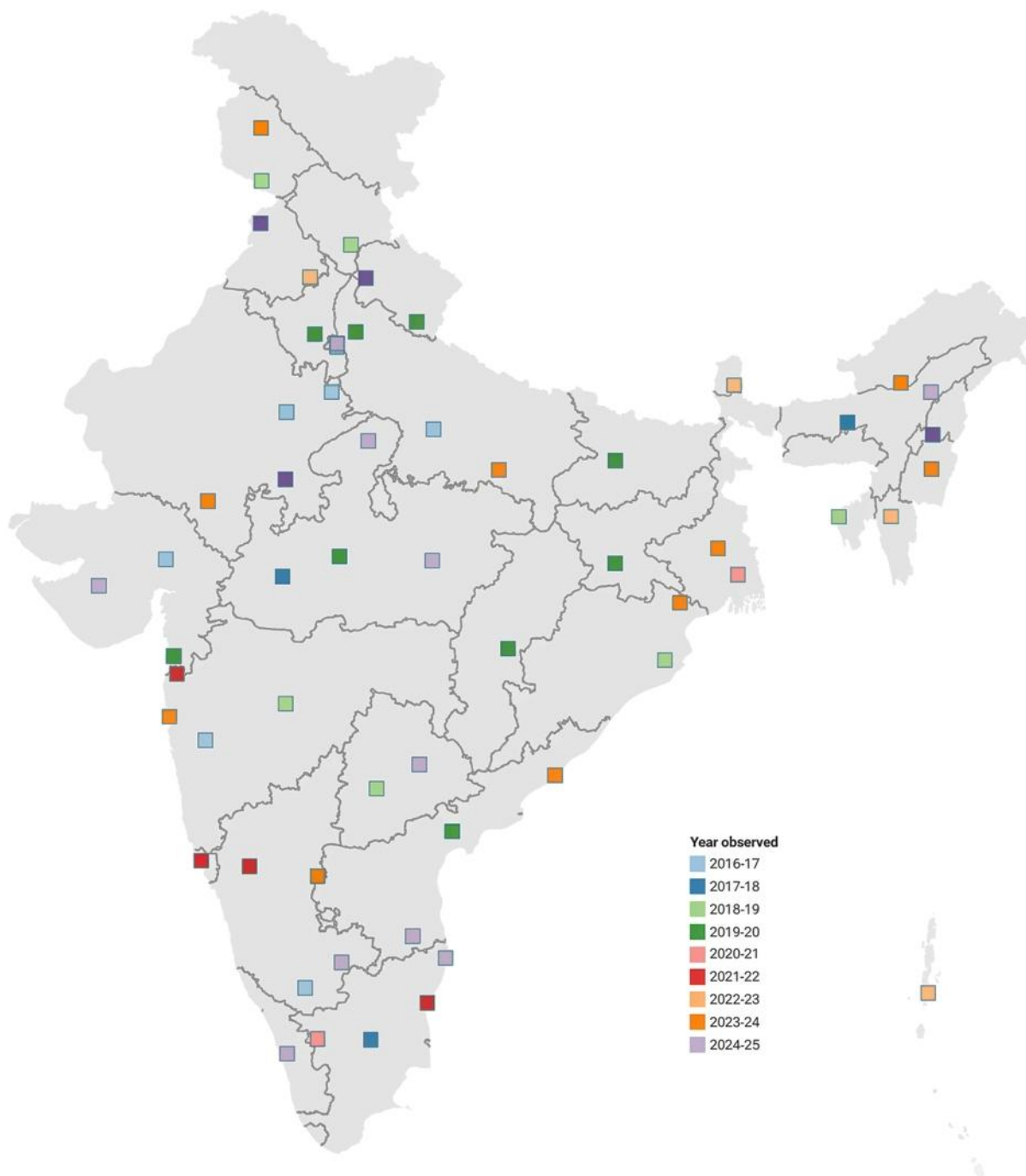
Specimen	<i>S. aureus</i>	<i>Enterococcus</i> spp.	<i>Klebsiella</i> spp.	<i>Escherichia coli</i>	<i>Acinetobacter</i> spp.	<i>Pseudomonas</i> spp.	<i>Salmonella</i> Typhi / Paratyphi	<i>Shigella</i> species	<i>Vibrio cholerae</i>
Blood	✓	✓	✓	✓	✓	✓	✓		
Urine		✓	✓	✓	✓	✓			
Pus Aspirate	✓	✓	✓	✓	✓	✓			
Other Sterile body fluids*	✓	✓	✓	✓	✓	✓			
Stool							✓	✓	✓

* Other sterile body fluids (OSBF)- Include abdominal fluid, amniotic fluid, bile, cerebrospinal fluid, cyst, endocardium, hip fluid, joint fluid, knee fluid, lymph node, semen, broncho-alveolar lavage, spleen, pleural fluid, pericardial fluid, bone marrow, Bartholin's cyst, fluid, gastric fluid, gallbladder, breast milk and prostatic fluid.

All NARS-Net sites submit AMR data of nine priority pathogens from the pre-defined specimen types to NCDC programme unit. Sites perform antimicrobial susceptibility testing by disk diffusion, broth microdilution, agar dilution, and automated antimicrobial susceptibility testing systems as per the programme SOPs. The programme unit at NCDC provides technical guidance and training support to the sentinel surveillance network laboratories towards accurate pathogen identification and antimicrobial susceptibility testing (AST), implementing quality management systems and standardized AMR surveillance data management.

² <https://ncdc.mohfw.gov.in/amr-semi-annual-bulletin/>

³ <https://ncdc.mohfw.gov.in/reports/>



Source: CBDDR, NCDC, MOHFW, GOI • Map data: © OSM • Created with Datawrapper

Figure 1: Geographical Location of National AMR Surveillance Network laboratories which have submitted AMR data from January to December 2025

National Reference Laboratory (NRL) has been established at NCDC in 2018-19 to implement External Quality Assessment Scheme (EQAS) for the network labs so as to ensure quality AMR Surveillance data. Additionally, NRL also confirms all the antibiotic-resistant alert strains, as defined under the programme, submitted by network sites and only the confirmed alerts are included in the annual report. Feedback on EQAS and Alert confirmation is shared with the respective sentinel sites. The NRL for AMR in fungal pathogens was established at VPCI,

Delhi University in the year 2021-22 for confirmation of all *Candida* spp. isolated from Blood culture specimens and provide trainings to programme sites.

The network sites are mandated to perform Internal quality control testing as per the programme SOPs.⁴ In addition, onsite support visits and trainings on data management are conducted to strengthen bacteriology laboratory capacity, identify the challenges and improve the quality of culture, identification and AST practices. The programme unit at NCDC also frequently organizes workshops on use of WHONET AMR data management software to facilitate timely AMR surveillance data flow from network sites to NCDC. Annual meetings to review the performance of each site under NARS-Net are also conducted to discuss and resolve site-specific challenges. This process plays a crucial role in strengthening and standardizing antimicrobial resistance surveillance across the network.

The WHONET 2024, an open-source offline microbiology data management desktop application, has been used for collecting, collating and analyzing routine antimicrobial susceptibility data generated by manual testing methods and automated systems at the laboratories. In this report, M100 34th Ed. of Clinical & Laboratory Standards Institute (CLSI) guideline⁵ is used for the classification of the isolates as susceptible (S), intermediate (I), or resistant (R). AMR data submitted monthly is reviewed during the monthly data quality monitoring virtual calls conducted by the designated AMR nodal officer at NCDC for the designated programme sites. The monthly data calls with the sites support them in validation of data with respect to the completeness of data fields and compliance with the AMR Surveillance panel of antibiotics. The errors observed in the data during data monitoring calls are notified to the sites in the feedback emails. The subsequently received revised corrected files are cleaned by NCDC nodal officers. A single file is generated from all the cumulative AMR data files from each site and data de-duplication is done before carrying out the analysis.

1.2. Findings

This annual AMR surveillance report includes data of 2,17,513 priority pathogen isolates submitted by 62 NARS-Net sites for the reporting period from 1st January 2025 to 31st December 2025 (List at Annexure-1.4).

All alert bacterial isolates (as per the SOP for Antimicrobial Resistance surveillance) included in the report have been confirmed at the NRL, NCDC. All the colistin-resistant Gram-negative priority bacterial isolates included in the report have been confirmed using broth microdilution method as per the SOP.

⁴ Standard Operating procedure for AMR Surveillance NARS-Net. January 2023. <https://ncdc.mohfw.gov.in/wp-content/uploads/2024/03/58495493521681880873.pdf>.

⁵ Clinical Laboratory Standard Institute. Performance Standard for Antimicrobial Susceptibility Testing. 34th Ed. CLSI Supplement M-100. Clinical and Laboratory Standard Institute. USA. 2024.

The revised WHONET data files received on monthly basis from 62 NARS-Net sites were combined to make a single file. The single file was then run through NCDC reporting MACROS in order to filter data of the priority pathogens and priority specimen types as per the programme data management SOP. Subsequently, the WHONET file was subjected to data deduplication (explained under section 1.2.1). Thereafter, the deduplicated WHONET data file was exported to excel format with the help of WHONET software for final data cleaning (Fig. 2). The cleaned data file in excel format was backlinked to obtain WHONET (.sqlite) file for initiating the data analysis and annual surveillance report preparation. The flow chart of data cleaning and data analysis is mentioned below:-

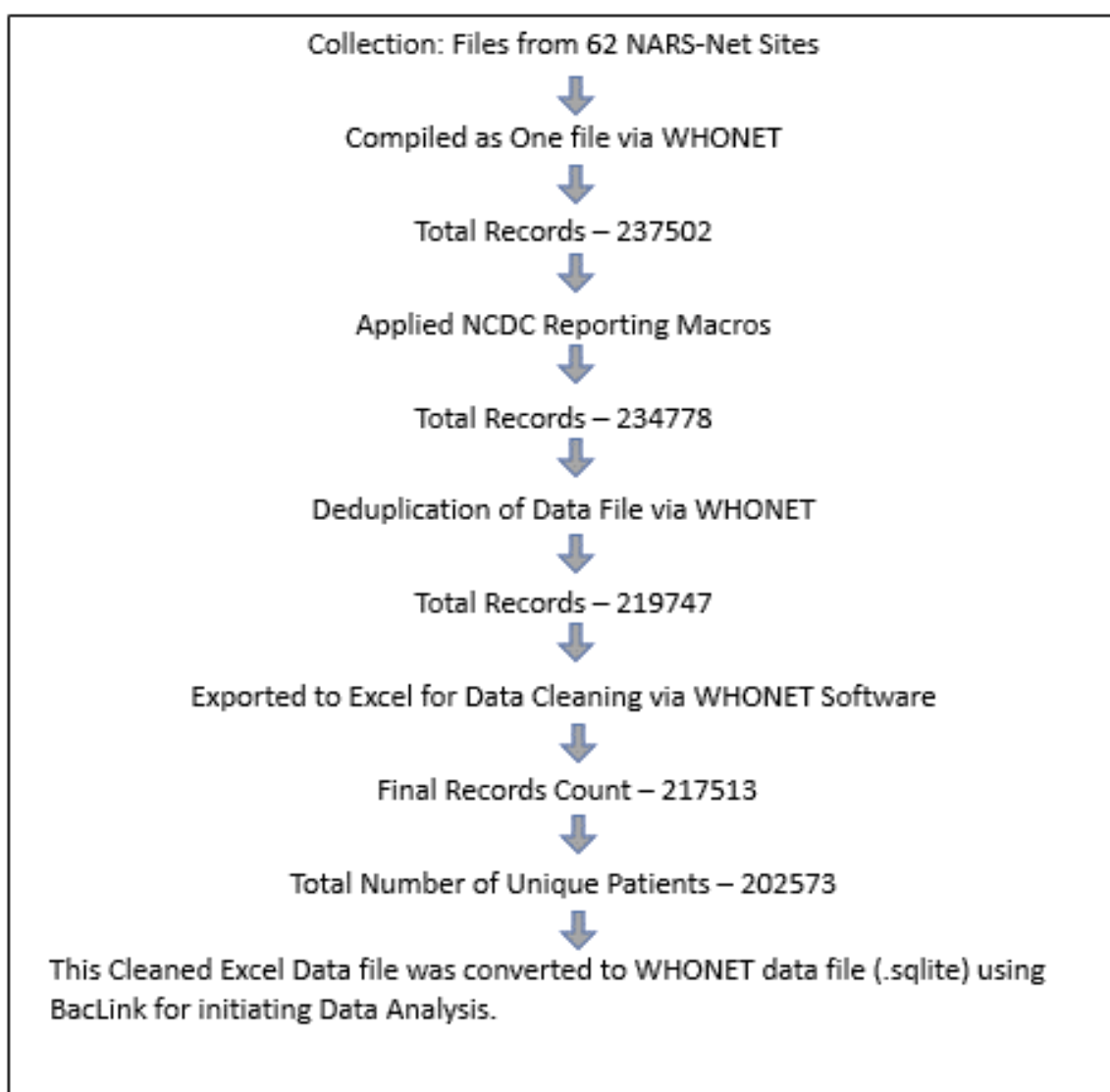


Figure 2: Flowchart to Obtain a Cleaned WHONET Data File for Analysis

1.2.1. Data Deduplication

For data analysis, only the first antibiotic susceptibility result has been considered for each patient per specimen type and pathogen. For example, if two blood cultures from the same patient yielded growth of *E. coli*, only the first has been included in the data; if the growth of *E. coli* was detected in one culture and of *K. pneumoniae* in the other, both results have been considered. If there was growth of *E. coli* in one blood culture and in one urine culture from the same patient, both specimen types have been included. If for the same patient, multiple isolates of same organism are included in the analysis, then there is a possibility that the data might get skewed towards resistance especially for isolates like *P.aeruginosa* and *A.baumannii*. Therefore, from each patient, only the first isolate of a given species from a particular specimen type isolated during the surveillance period was included, regardless of its susceptibility profile.

Deduplication of reported data of 2,17,513 isolates using WHONET revealed 2,02,573 unique patients. For the surveillance period, only one result has been considered for each patient per specimen type and each pathogen. The deduplicated data file has been used for AST analysis.

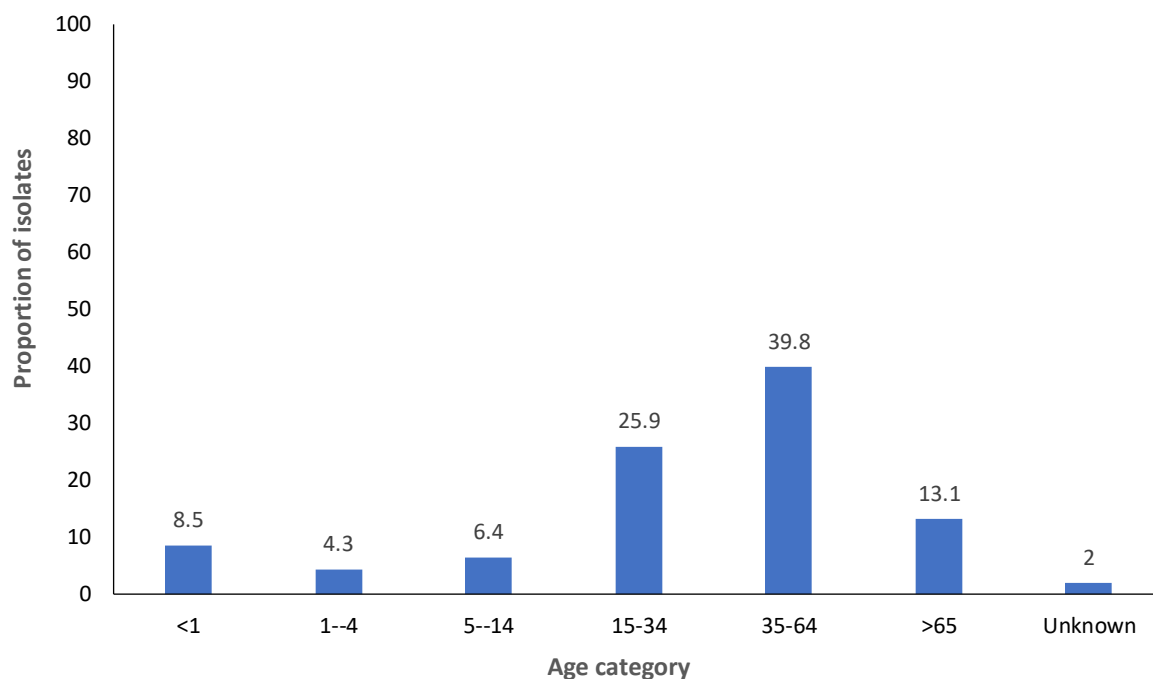
1.2.2. Age and Gender distribution of reported AMR data

Among 2,17,513 unique patients, AST data in this report is from 52.4% of male and 47.5% of female patients which is similar to the findings of last year (Fig 3). Majority of patients were in the age category 35-64 years (39.8%) and 4.3% patients were in the age category 1-4 years. About 65.7 % patients were in the productive age group of 15-64 years. Figure 4 represents the distribution of priority bacterial pathogen isolates by age category.



*Gender details were missing for 244 (0.1%) isolates

Figure 3: Distribution of unique patients according to gender (N=2,02,573)



*Age details were missing for 4055 (2%) isolates

Figure 4: Distribution of priority pathogen isolates by age category (N=2,17,513)

The analysis of 2025 AMR surveillance data revealed that the highest number of isolates were from urine specimens (44.5%) (Table 2). The most commonly isolated priority bacterial pathogen was *E. coli* (31%), which is similar to the previous five years, followed by *Klebsiella* species (24%), *Pseudomonas* species (12%), *S. aureus* (11%), *Acinetobacter* species (10%) *Enterococcus* species (10%) and *Salmonella* enterica serovar *Typhi* and *Paratyphi* (0.54%) (Table 3).

Table 2: Distribution of priority pathogen isolates based on specimen type (N=2,17,513)

Specimen Type	Number of Isolates	%
Urine	96742	44.5
Blood	38238	17.6
Pus Aspirate	68281	31.4
Other sterile body fluids	13995	6.4
Stool	257	0.1
Total	2,17,513	100

E. coli (49%) was the most commonly isolated pathogen from urine specimens; *Klebsiella* species (25%) was the predominant isolate from blood specimens and from pus specimens *Klebsiella* spp (24%) and *S. aureus* (23%) were the predominant isolates. Among stool specimens, most common pathogen reported was *Shigella* species (46.6%), followed by *Vibrio cholerae* (41.3%), and *Salmonella Typhi & Paratyphi* (12.1%) (Table 3). Among the non-fermenting Gram-negative bacilli, *Pseudomonas* species was the most commonly isolated pathogen from pus aspirates (19%) and *Acinetobacter* species from blood culture specimens (23%) (Table 4). Amongst Gram-positive organisms, *Enterococcus* species was most the commonly isolated pathogen from urine specimens (68%) and *S. aureus* from pus aspirates (67%) (Table 4).

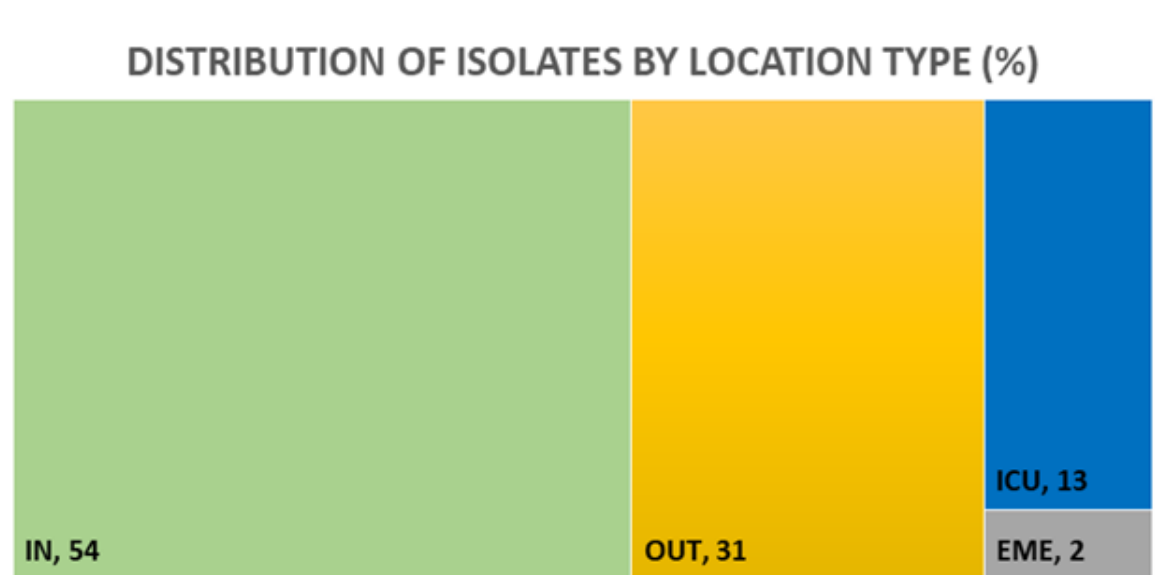
Table 3: Specimen type wise distribution of isolates (N= 2,17,513)

Priority Pathogen	Blood		Pus aspirate		OSBF		Urine		Stool	Total	
	N	(%)	N	(%)	N	(%)	N	(%)	N	N	(%)
<i>S. aureus</i>	6775	(18)	15677	(23)	1055	(8)	x		X	23507	(11)
<i>Enterococcus</i> spp.	3677	(10)	2320	(3)	1055	(8)	15266	(16)	X	22318	(10)
<i>Escherichia Coli</i>	3795	(10)	14362	(21)	2958	(21)	47226	(49)	X	68341	(31)
<i>Klebsiella</i> spp.	9478	(25)	16379	(24)	3586	(26)	23154	(24)	X	52597	(24)
<i>Salmonella Typhi</i> and <i>Paratyphi</i>	1145	(3)	x		x		x		31 (12.1)	1176	(0.54)
<i>Pseudomonas</i> spp.	4478	(12)	12739	(19)	2369	(17)	7051	(7)	X	26637	(12)
<i>Acinetobacter</i> spp.	8890	(23)	6804	(10)	2972	(21)	4045	(4)	X	22711	(10)
<i>Shigella</i> species	x		x		x		x		120 (46.6)	120	(0.05)
<i>Vibrio cholerae</i>	x		x		x		x		106 (41.3)	106	(0.04)
Total	38238	(100)	68281	(100)	13995	(100)	96742	(100)	257 (100)	217513	(100)

x - Specimen type not included under the program; *OSBF- Include abdominal fluid, amniotic fluid, bile, cerebrospinal fluid, cyst, endocardium, hip fluid, joint fluid, knee fluid, lymph node, semen, broncho-alveolar lavage, spleen, pleural fluid, pericardial fluid, bone marrow, bartholin's cyst, fluid, gastric fluid, gall bladder, breast milk and prostatic fluid

Table 4: Proportion of priority pathogens isolated from various specimen types (N= 2,17,513)

Priority	<i>S. aureus</i>	<i>Enterococcus</i> spp.	<i>E. coli</i>	<i>Klebsiella</i> spp.	<i>S. Typhi</i> and <i>Paratyphi</i>	<i>Pseudomonas</i> spp.	<i>Acinetobacter</i> spp.	<i>Shigella</i> spp.	<i>Vibrio cholerae</i>
	N (%)	N (%)	N (%)	N (%)	N (%)	N (%)	N (%)	N	N
Blood	6775 (29)	3677 (17)	3795 (6)	9478 (18)	1145 (98)	4478 (16)	8890 (39)	x	x
Pus aspirate	15677 (67)	2320 (10)	14362 (21)	16379 (31)	x	12739 (48)	6804 (30)	x	x
OSBF	1055 (4)	1055 (5)	2958 (4)	3586 (7)	x	2369 (8)	2972 (12)	x	x
Urine	x	15266 (68)	47226 (69)	23154 (44)	x	7051 (28)	4045 (18)	x	x
Stool	x	x	x	x	31 (2)	x	x	120	106
Total	23507 (100)	22318 (100)	68341 (100)	52597 (100)	1176 (100)	26637 (100)	22711 (100)		



IN – Inpatient; OUT–Outpatient; ICU –Intensive care; EME –Emergency; UNK – Unknown; *Location type field was missing in 17 isolates

Figure 5: Distribution of isolates by location type (N=217513)

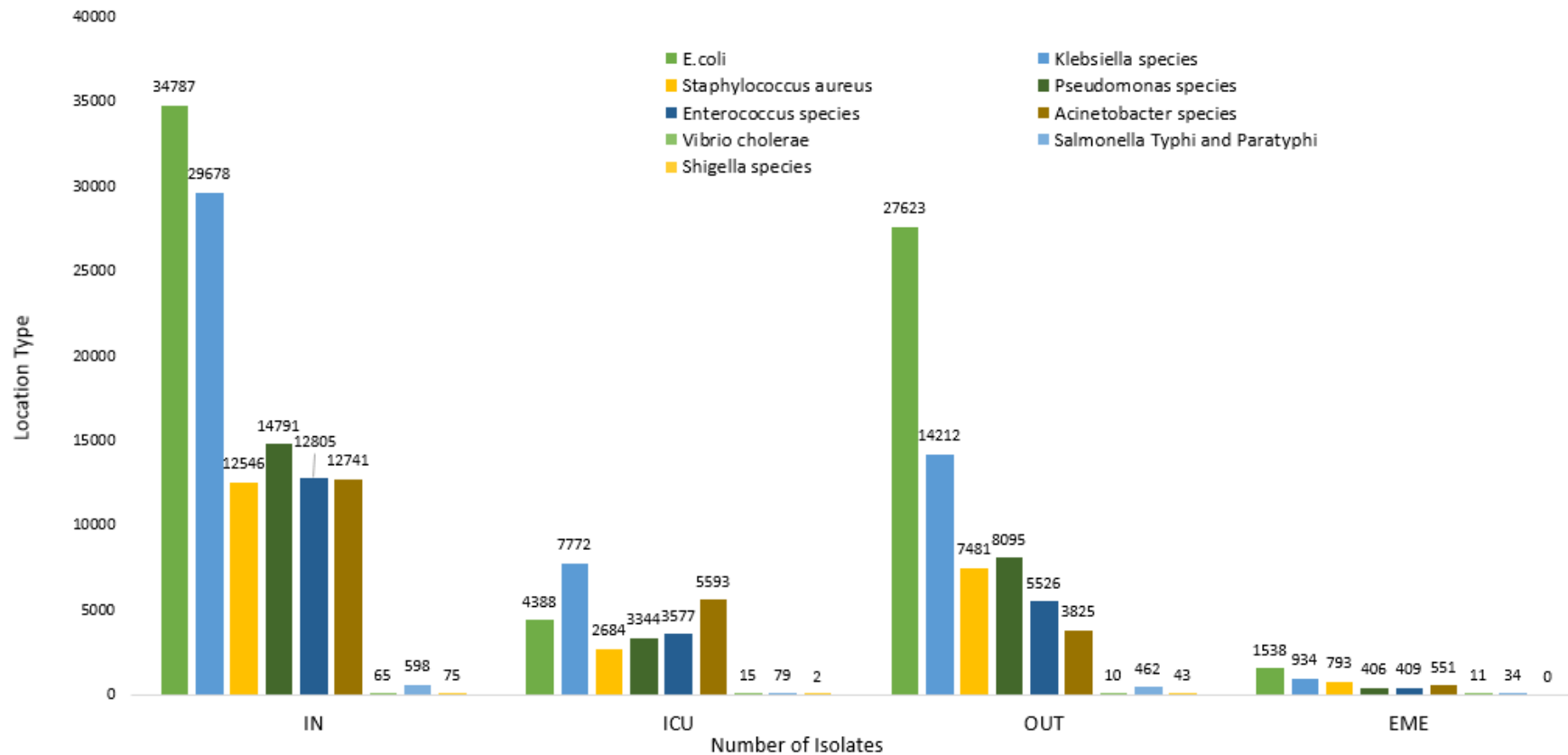
In the 2025 AMR surveillance data, the majority of isolates (54%) were from patients admitted in hospital wards whereas the least number of isolates (2%) belonged to patients from the Emergency department. Almost a third of the isolates (31%) were from patients visiting the outpatient clinics and 13% of the priority pathogens were isolated from Intensive care units (ICUs) (Fig. 5).

Table 5: Distribution of priority pathogen isolates by location type (N=217513)

Priority Pathogen	Inpatient (N=118086)		Outpatient (N=67277)		I.C.U. (N=27454)		Emergency (N=4676)	
	N	(%)	N	(%)	N	(%)	N	(%)
<i>Escherichia coli</i>	34787	(29)	27623	(41)	4388	(16)	1538	(33)
<i>Klebsiella</i> species	29678	(25)	14212	(21)	7772	(28)	934	(20)
<i>Salmonella Typhi</i> and <i>Paratyphi</i>	598	(0.5)	462	(0.7)	79	(0.3)	34	(0.7)
<i>Pseudomonas</i> species	14791	(13)	8095	(12)	3344	(1)	406	(9)
<i>Acinetobacter</i> species	12741	(11)	3825	(6)	5593	(20)	551	(12)
<i>S. aureus</i>	12546	(11)	7481	(11)	2684	(10)	793	(17)
<i>Enterococcus</i> species	12805	(11)	5526	(8)	3577	(13)	409	(9)
<i>Shigella</i> species	75	(0.06)	43	(0.06)	02	(0.01)	00	(0)
<i>Vibrio cholerae</i>	65	(0.06)	10	(0.01)	15	(0.1)	11	(0.24)

*Location type field was missing in 20 isolates

Among the inpatients, the most commonly isolated priority pathogen was *Escherichia coli* (29%) followed by *Klebsiella* spp. (25%), whereas in Intensive care units *Klebsiella* spp. (28%) was the most commonly isolated pathogen followed by *Acinetobacter* spp. (20%) and *Escherichia coli* (16%) (Table 5). *Escherichia coli* was also the most commonly isolated pathogen from outpatient clinics (41%) and emergency departments (33%) (Table 5, Fig. 6).



Location type field was missing in 20 isolates

Figure 6: Distribution of priority pathogen isolates by location-type

1.2.3. AMR Surveillance of priority pathogens

1.2.3.1. Gram-Positive Cocci

The AMR surveillance under NARS-Net covers the two most prevalent Gram-positive bacterial human pathogens i.e., *Staphylococcus aureus* and *Enterococcus* species.

Staphylococcus aureus

During the 2025 reporting period, a total of 23,507 *S. aureus* isolates data was submitted to NCDC (Table-3) constituting 11% of priority pathogens included in this annual report. AST data analysis of 23,507 isolates revealed that majority of *S. aureus* were isolated from pus specimens (67%). In this report, *S. aureus* was observed to be a significant cause of bacteremia (29%) (Table 4).

Approximately more than half of *S. aureus* isolated from blood (61%; 95% CI: 60.1-62.7) were resistant to ceftazidime (a surrogate for mecA-mediated oxacillin resistance), while the resistance to ceftazidime in pus aspirates (55%; 95% CI: 54.0-55.7) and other sterile body fluids (55%; CI: 51.7-58.3) was lower in comparison to the blood isolates (Table 6). For analysis of linezolid resistance in *S. aureus*, CLSI breakpoint for 2025 have been considered. Resistance to Linezolid in *S. aureus* isolated from blood was found to be less than 1%, consistent with the observations over last three years. Out of 14,833 isolates tested on Vancomycin screen agar, no isolates showed growth on the vancomycin screen agar plate, hence no resistance was observed to vancomycin.

Among the AMR Surveillance priority pathogens isolated in the data reporting period, *S. aureus* was mainly isolated from inpatient wards followed by outpatient clinics and intensive care settings (Table 5).

Among 11,090 isolates of MRSA all were sensitive to vancomycin and 7 isolates from blood cultures were resistant to linezolid. All linezolid resistant isolates were confirmed at AMR-NRL NCDC. Proportion of MRSA in blood has slightly increased from last year in all location settings, from 63% to 65% in intensive care units, from 55% to 61% amongst inpatients and from 44% to 51% in outpatients (Fig. 7).

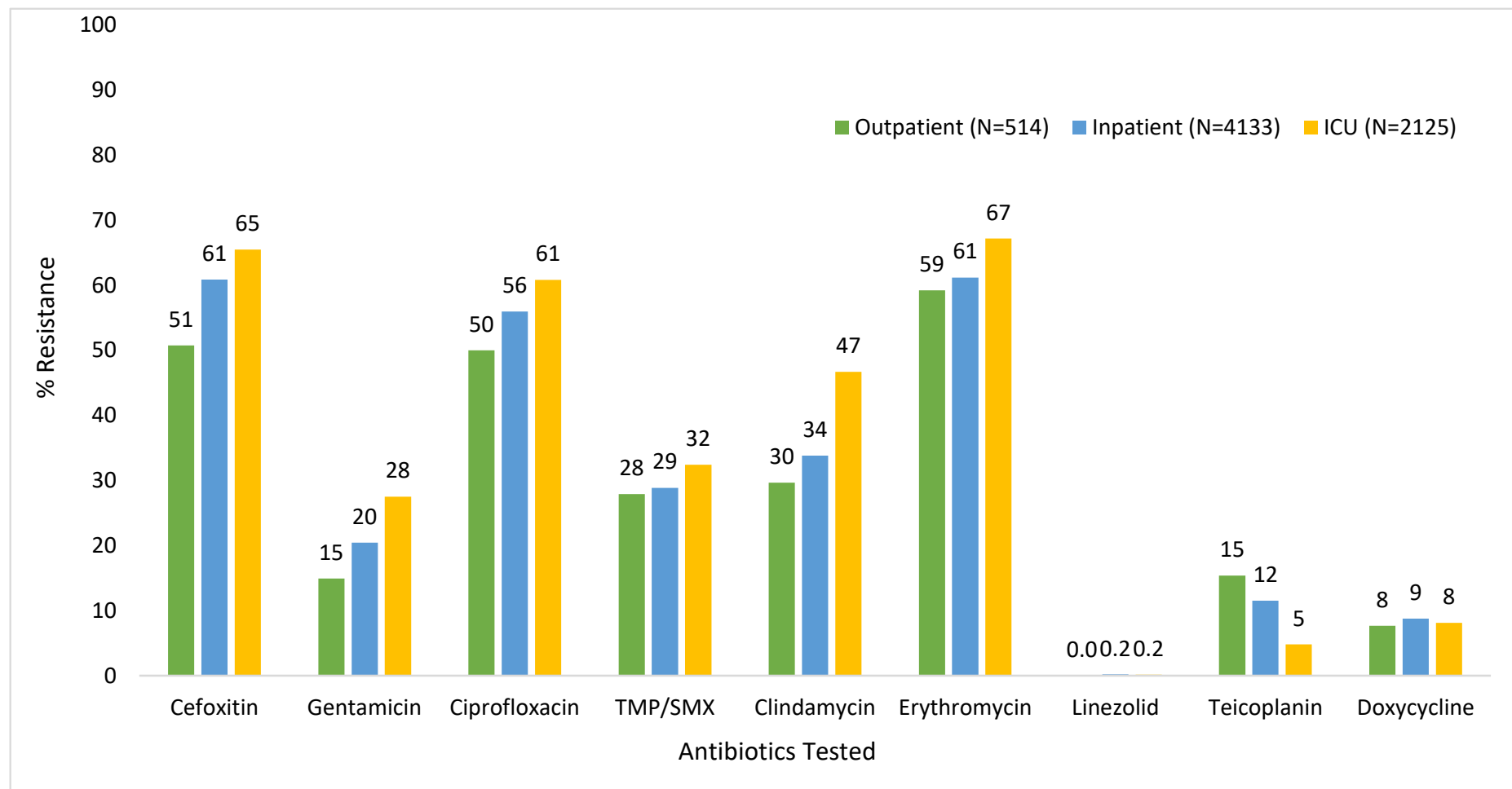
Resistance to all the surveillance panel antibiotics was observed to be higher among isolates from intensive care settings than those from outpatient departments and the inpatient wards (Fig. 7).

Table 6: Resistance profile of *Staphylococcus aureus* (N=23,507)

Antibiotic tested	Blood (N=6,775)			Pus Aspirate (N=15,677)			Other Sterile Body Fluids (N=1,055)		
	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI
Cefoxitin	5538	61	60.1-62.7	13135	55	54.0-55.7	876	55	51.7-58.3
Ciprofloxacin	5407	58	56.6-59.2	12119	75	73.9-75.5	860	58	55.0-61.7
Clindamycin	6038	37	36.2-38.7	14471	22	20.9-22.2	961	31	27.9-33.8
Doxycycline	5205	8	7.8-9.3	10445	4	3.8-4.6	746	6	4.5-8.1
Erythromycin	6354	63	61.9-64.3	14816	53	52.2-53.8	965	55	51.7-58.1
Gentamicin	5604	22	20.9-23.1	12763	22	21.0-22.5	899	19	16.1-21.3
Linezolid	6375	0.2	0.1-0.3	14324	0.2	0.1-0.3	985	0.3	0.1-1.0
TMP/SMX	5311	30	28.9-31.4	12522	21	20.3-21.7	862	25	22.3-28.2
Teicoplanin	1178	9	7.7-11.1	2505	6	5.3-7.2	243	6	3.6-10.2

Alert pathogens confirmed at NRL have been included in the data

TMP-SMX: Trimethoprim-Sulfamethoxazole



*Data of the emergency department was clubbed with data from inpatient wards; Location type for 3 isolate data is unknown

TMP-SMX: Trimethoprim-Sulfamethoxazole

Figure 7: Resistance profile of *S. aureus* in blood (N=6775) by location type

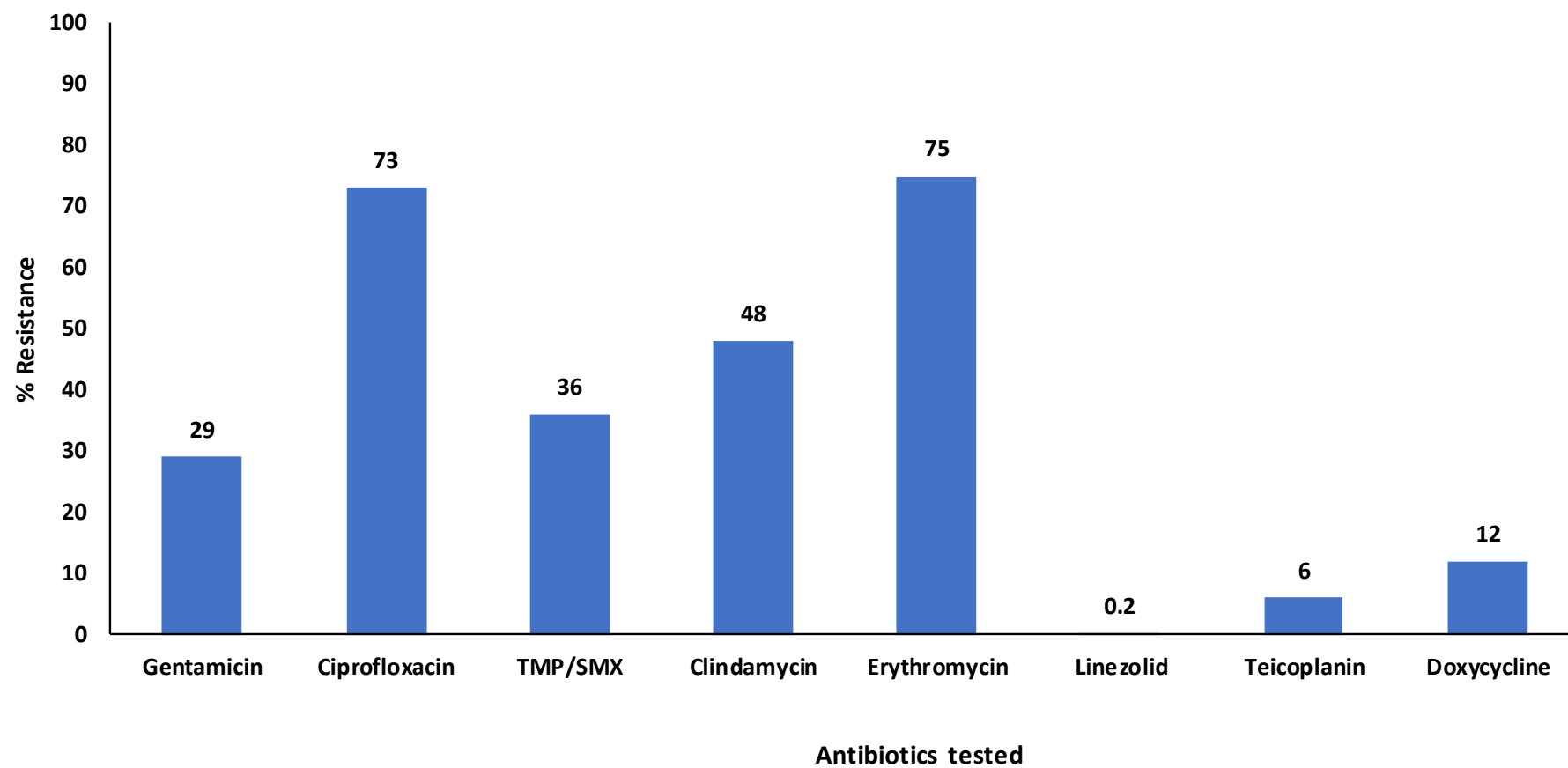


Figure 8: Resistance profile of Methicillin Resistant *S. aureus* (MRSA) in blood (N=3,251)

Enterococcus species

Enterococcus species isolates comprised of 10% of the annual AMR data and contributed to 49% of the Gram-positive cocci data reported during 2025 (Table 3). A total of 22,318 *Enterococcus* species isolates data was submitted by the NARS-Net sites (Fig. 2). Upon analysis, isolation rates from specimen types, namely urine, blood, pus aspirates and other sterile body fluids were 68%, 17%, 10% and 5% respectively (Table 4). Among *Enterococcus* species isolated from blood, highest resistance was observed to erythromycin 79% (CI: 77.8-80.6) followed by Ampicillin (69%, CI: 67.0-70.3) (Table 7). Notably resistance to vancomycin (30%, CI: 28.1-31.8) has increased from previous three years while the resistance to teicoplanin (24%, CI: 22.0-25.2) has slightly decreased from last year (26%). Resistance to linezolid (2.6%, CI: 2.0-3.2) has slightly increased among blood isolates from last year's report (1.7%).

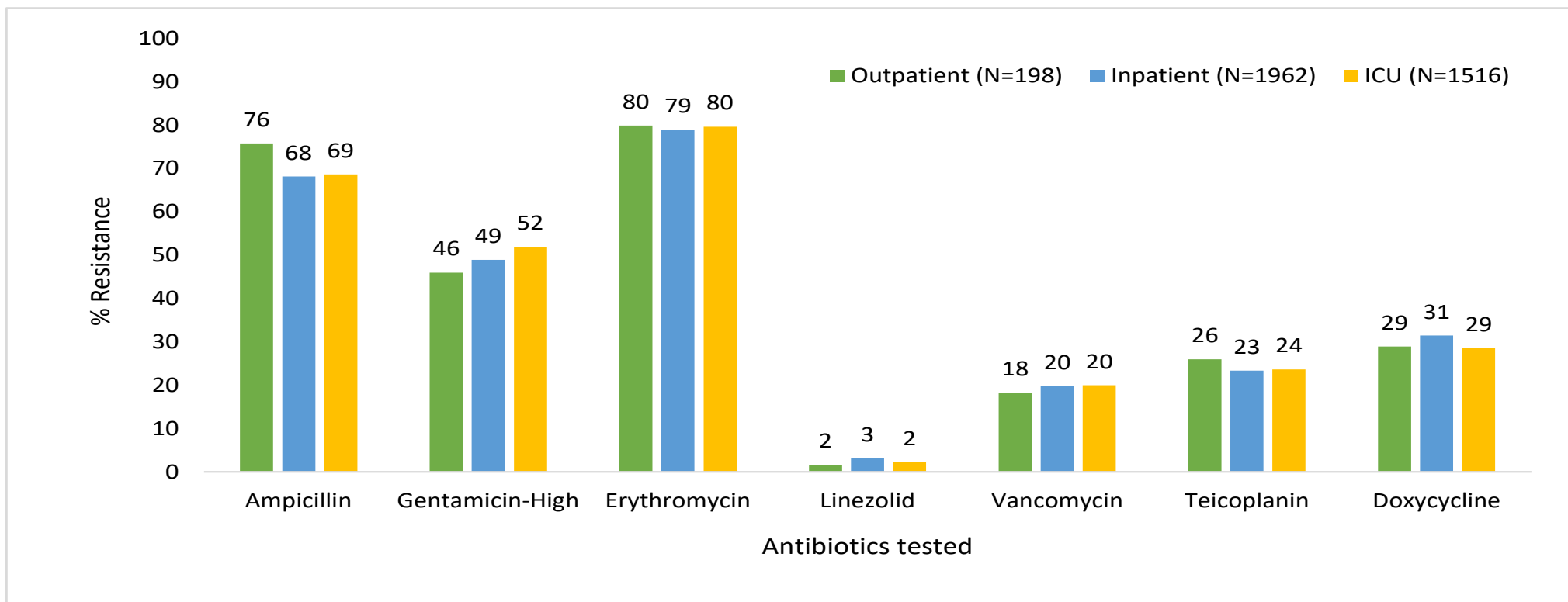
Among the urinary isolates, 58% were resistant to first-line antibiotic ampicillin (CI: 56.7-58.4) and 50% to gentamicin-high (CI: 48.7-58.4). Highest resistance was observed to ciprofloxacin (81%, CI: 80.62-82.0) and lowest was observed to linezolid 0.7 % (CI: 0.6-0.8). Resistance to vancomycin among urinary isolates has decreased to 9% in 2025 from 12% in 2024 (Table 7).

Majority of *Enterococcus* spp. isolates were from patients in IPD followed by OPD and ICU (Table 5). AST analysis of blood isolates of *Enterococcus* spp. showed low level resistance to linezolid in isolates from patients from IPD, ICUs and OPD (Fig. 9). The incidence of VRE among blood isolates was 20% in both inpatients and ICU settings which has slightly decreased from the previous year (22%) (Fig. 9). Higher vancomycin resistance was observed in blood isolates (30%) followed by other sterile body fluids, urine and pus aspirates (Fig. 9, 10). Among urinary isolates of *Enterococcus* species, 45% resistance was observed to nitrofurantoin in ICU settings.

Table 7: Resistance profile of *Enterococcus* species (N=22,318)

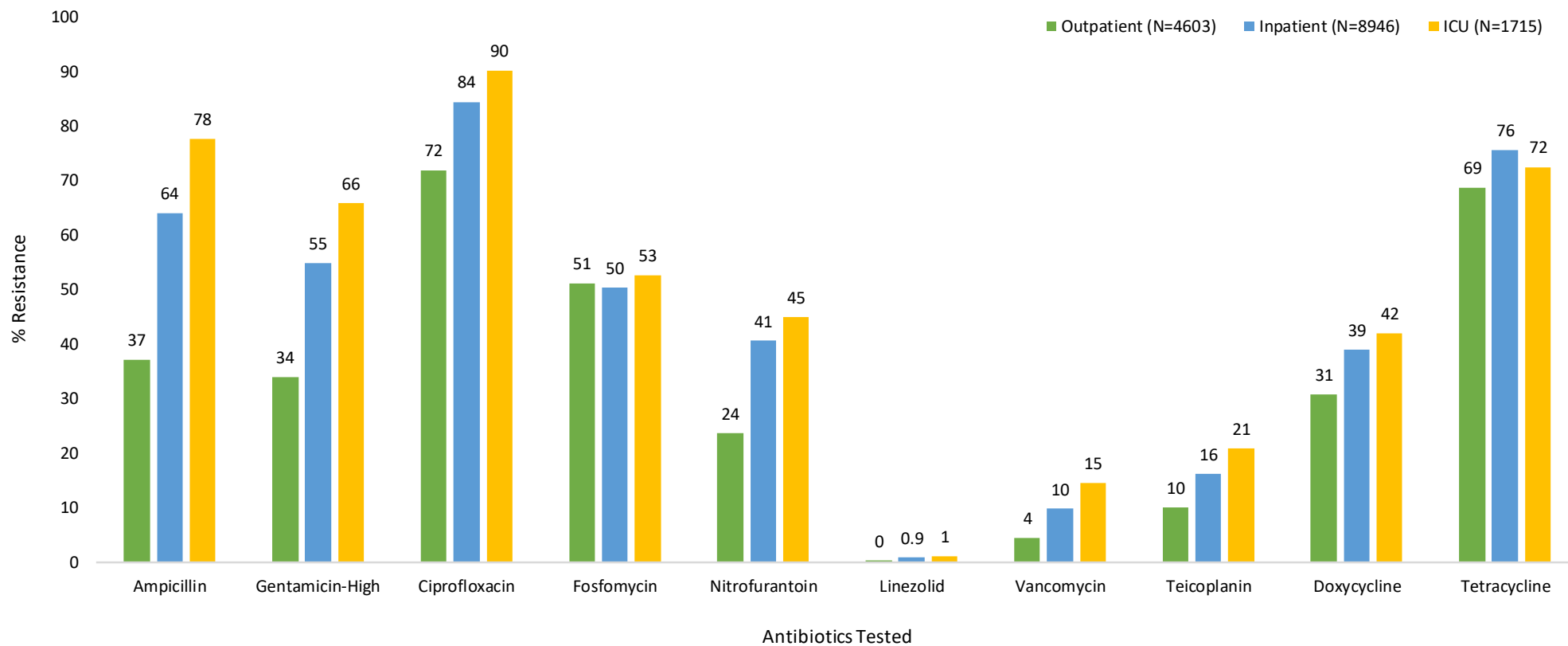
Antibiotic Tested	Blood (N=3677)			Pus Aspirate (N=2320)			OSBF (N=1055)			Urine (N=15,266)		
	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI
Ampicillin	2969	69	67.0-70.3	1745	47	44.2-48.9	724	62	58.2-65.4	12111	58	56.7-58.4
Doxycycline	3130	50	48.2-51.7	1478	29	26.7-31.4	668	31	27.4-34.5	4676	37	36.0-38.8
Erythromycin	3218	79	77.8-80.6	1970	73	71.4-75.3	895	73	69.7-75.6	x		
Gentamicin-High	3433	3	2.2-3.3	1926	32	30.1-34.3	922	35	32.3-38.6	12948	50	48.7-50.4
Linezolid	2999	2.6	2.0-3.2	2111	1	0.8-1.8	978	2	1.0-2.7	14511	0.7	0.6-0.8
Teicoplanin	2648	24	22.0-25.2	1565	17	15.0-18.8	768	19	16.1-21.7	9842	15	14.3-15.7
Vancomycin	2437	30	28.1-31.8	2069	8	7.0-9.4	970	14	12.2-16.7	14448	9	8.2-9.1
Ciprofloxacin	x			x			x			11980	81	80.6-82.0
Fosfomycin	x			x			x			7350	51	49.7-52.0
Nitrofurantoin	x			x			x			13273	36	35.1-36.8
Tetracycline	x			x			x			8420	73	72.1-74.0

*Alert pathogens confirmed at NRL, NCDC only were included in the data



*Data from the emergency department was clubbed with data from inpatient wards; the Location type for 1 isolate was unknown.

Figure 9: Resistance profile of *Enterococcus* species in blood (N=3,677) by location type



*Data from the emergency department was clubbed with data from inpatient wards; the Location type for 2 isolates is unknown.

Figure 10: Resistance profile of *Enterococcus* species in urine (N=15,266) by location type

1.2.3.2. Gram-Negative Bacilli

Seven Gram-Negative bacteria of public health importance are included under AMR surveillance being coordinated at NARS-Net sites. These are *Escherichia coli*, *Klebsiella* species, *Pseudomonas* species, *Acinetobacter* species, *Salmonella* enterica serovar Typhi and Paratyphi, *Shigella* species and *Vibrio cholerae*.

1.2.3.2.1. Enterobacteriaceae

Enterobacteriaceae accounted for 56% of all the priority pathogens.

Escherichia coli

E. coli (68,341) contributed to one-third of the total isolates data during the year 2025. *E. coli* was predominantly isolated from the urine specimens (69%) followed by pus aspirates (21%), blood (6%) and other sterile body fluids (4%) respectively (Table 4).

E. coli isolates showed more than 50% resistance to all the tested antibiotics except carbapenems, colistin and aminoglycosides. The highest resistance was observed to ampicillin as noted in previous years (Table 8). High resistance to ciprofloxacin was observed ranging 73% to 80% in all specimen types. Resistance to nitrofurantoin in urinary isolates showed an increasing trend over last 4 years (increased from 9% in 2022 to 22% in 2025) (Table 8).

Among third generation cephalosporins tested, resistance ranging between 79% to 85% was observed in isolates from all specimen types. β -lactam- β -lactamase inhibitor combination i.e. amoxicillin-clavulanate also had high resistance of 67% in blood isolates. Among carbapenems, ertapenem (46%) had higher resistance than imipenem (39%) and meropenem (38%) in *E. coli* blood isolates (Table 8). For colistin susceptibility testing, isolates were tested using colistin agar dilution (ref: CLSI-M100 Ed. 35) and broth microdilution method, breakpoints as per EUCAST guidelines were used. Among all the colistin resistant isolates submitted by the NARS-Net sites, two isolates from blood (0.08%), one from OSBF (0.06%), two from pus aspirates (0.02%) and thirteen from urine (0.05%) were confirmed as colistin resistant at AMR-NRL, NCDC.

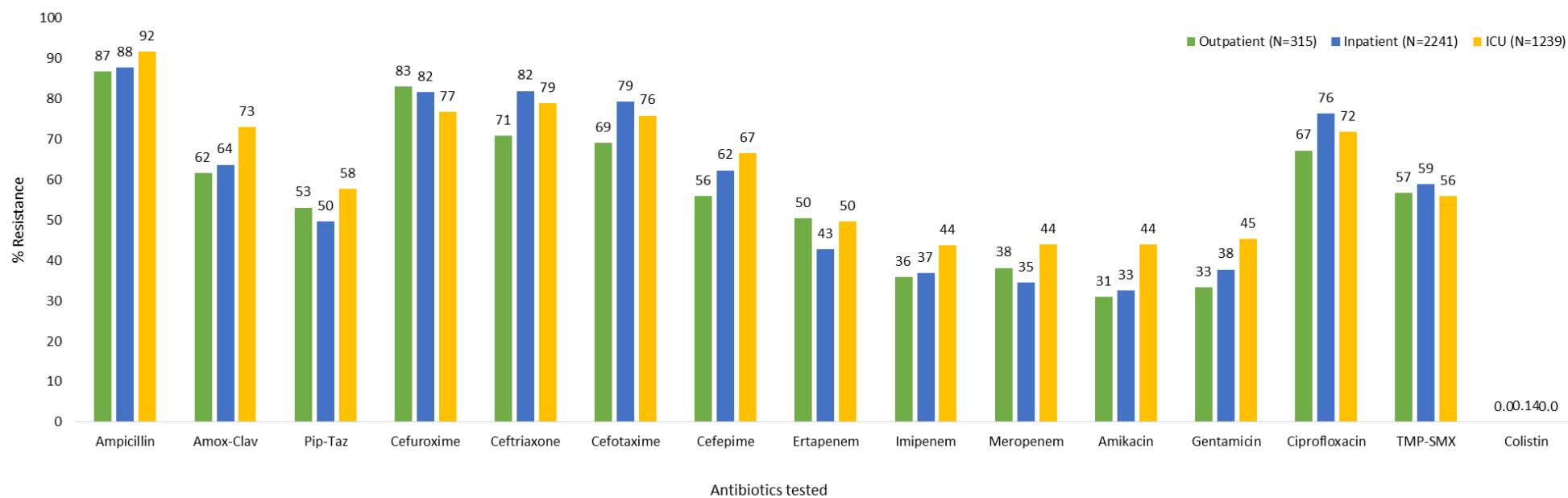
Among the blood isolates of *E. coli*, resistance profile based on the location type showed higher resistance to all the antibiotics in the surveillance panel in isolates from the intensive care units as compared to isolates from inpatient and outpatient departments (Fig. 11). Among urinary isolates of ICU patients, 23% resistance was observed to nitrofurantoin. (Fig. 12) Carbapenemase producing *E. coli* isolates were most frequently isolated from blood, followed by pus and OSBF and least from urine specimens of patients in all location types (Fig 11; 12).

Table 8: Resistance profile of *Escherichia coli* (N=68,341)

Antibiotic Tested	Blood (N=3,795)			PA (N=14,362)			OSBF (N=2,958)			Urine (N=47,226)		
	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI
Ampicillin	2313	89	87.8-90.4	8640	89	88.8-90.1	1759	91	89.0-91.8	33473	88	87.5-88.2
Amox-Clav	3003	67	65.0-68.4	11212	67	65.6-67.4	2218	68	66.3-70.2	34903	62	61.1-62.1
Pip-Taz	3436	53	51.0-54.3	12502	49	48.0-49.7	2536	56	54.0-57.9	39468	42	41.9-42.8
Cefuroxime	2030	80	78.3-81.8	7685	80	78.7-80.5	1375	86	84.0-87.7	23400	74	73.6-74.7
Ceftriaxone	2696	80	78.7-81.7	9057	78	77.2-78.9	1887	83	81.3-84.7	26628	72	71.3-72.4
Cefotaxime	1902	77	75.4-79.2	9293	79	77.8-79.4	1662	81	79.0-82.8	35029	74	73.7-74.6
Cefepime	3197	63	61.6-64.9	11176	60	59.1-60.9	2413	61	59.0-62.9	32263	53	52.3-53.4
Ertapenem	2000	46	43.5-47.9	7506	36	34.9-37.1	1481	45	42.4-47.5	20851	25	24.6-25.8
Imipenem	3192	39	37.5-40.9	11823	31	30.0-31.7	2349	36	34.1-38.0	34533	24	23.9-24.8
Meropenem	2954	38	36.3-39.8	11585	29	28.3-30.0	2212	37	34.6-38.6	30262	22	21.4-22.3
Amikacin	3512	36	34.7-37.9	12587	28	27.4-29.0	2567	29	27.0-30.5	38720	29	28.5-29.4
Gentamicin	3258	40	38.3-41.7	11409	34	33.2-35.0	2371	34	32.0-35.9	37833	33	32.2-33.2
Ciprofloxacin	3230	74	72.6-75.6	11789	76	75.3-76.8	2377	78	76.3-79.7	37295	73	73.0-73.9
TMP-SMX	3085	58	56.0-59.5	11332	57	56.3-58.1	2334	57	55.5-59.5	39173	54	53.5-54.5
Colistin	2531	0.08	0-0.3	8304	0.02	0-0.1	1776	0.06	0-0.4	26119	0.05	0-0.1
Fosfomycin		X			X			X		28059	5	4.8-5.3
Nitrofurantoin		X			X			X		43497	22	21.6-22.4
Doxycycline		X		3882	42	40.7-43.9	1006	48	44.6-50.9		X	

x- Drug bug combination for the specimen type not included in the NARS-Net surveillance panel

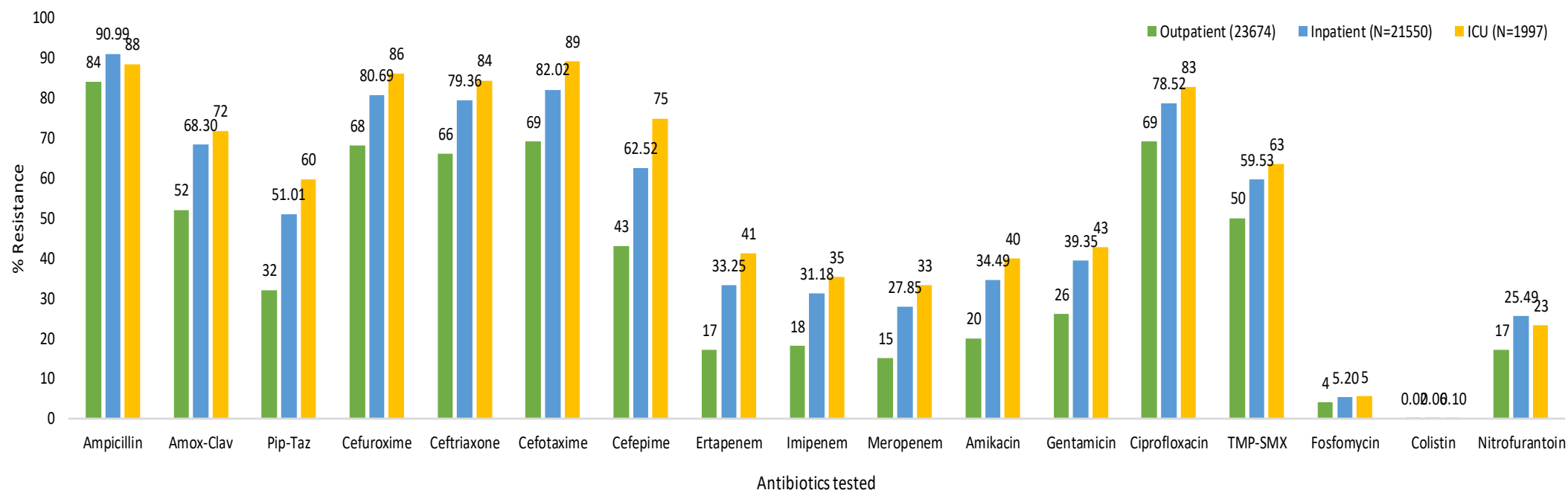
TMP/SMX - Trimethoprim/sulfamethoxazole
Amox-clav -Amoxicillin/Clavulanic acid



*Data from the emergency department was clubbed with data from inpatient wards.

Pip-Taz: Piperacillin-Tazobactam
 Amox-Clav: Amoxicillin-Clavulanic acid
 TMP-SMX: Trimethoprim-Sulfamethoxazole

Figure 11: Resistance profile of *Escherichia coli* isolated from blood (N=3,795) by location type



Pip-Taz: Piperacillin-Tazobactam, Amox-Clav: Amoxicillin-Clavulanic acid, and TMP-SMX: Trimethoprim-Sulfamethoxazole

*Data from the emergency department was clubbed with data from inpatient wards; the Location type for 5 isolates was unknown.

Figure 12: Resistance profile of *Escherichia coli* isolated from urine (N=47,226) by location type

Klebsiella species

In the current data reporting period 52,597 isolates of *Klebsiella* species were reported. The isolation of *Klebsiella* spp. in data reported by the sentinel sites was highest from urine specimens (44%) followed by pus aspirate (31%), blood (18%) and OSBF (7%) (Table 4).

Blood isolates showed more than 50% resistance to all tested antibiotics except for colistin. ESBL producing *Klebsiella* species isolated from blood were higher (84%) in number than from urine specimens (69%). Majority of carbapenemase producing *Klebsiella* species were isolated from blood (58%-69%) followed by pus aspirates (46-53%), OSBF (46%-54%), and urine specimens (36% - 42%). Forty five percent of the urinary isolates showed resistance to aminoglycosides (amikacin and gentamycin); whereas more than half of the blood isolates were resistant to aminoglycosides (amikacin; 63% and gentamicin; 56%) (Table 9). Approx. 58% of urinary isolates showed resistance to nitrofurantoin.

Resistance to the reserve antibiotic, colistin, was found to be highest in *Klebsiella* spp. compared to other Gram-negative priority pathogens isolated from all the specimen types which is similar to observations in previous year's report. All colistin resistant isolates included in this report have been confirmed at the AMR NRL at NCDC.

Location type wise AST data revealed higher resistance to all the tested antibiotics in *Klebsiella* species isolated from blood of ICU patients compared to the *Klebsiella* species isolated from inpatients and outpatients (Fig. 13). Among carbapenems, ertapenem showed highest resistance (72%) followed by meropenem (64%) and imipenem (64%) in blood isolates from ICU patients. Similarly, high resistance was observed to third generation cephalosporins viz. cefotaxime (86%), cefuroxime (87%) and ceftriaxone (87%) among blood isolates from intensive care units.

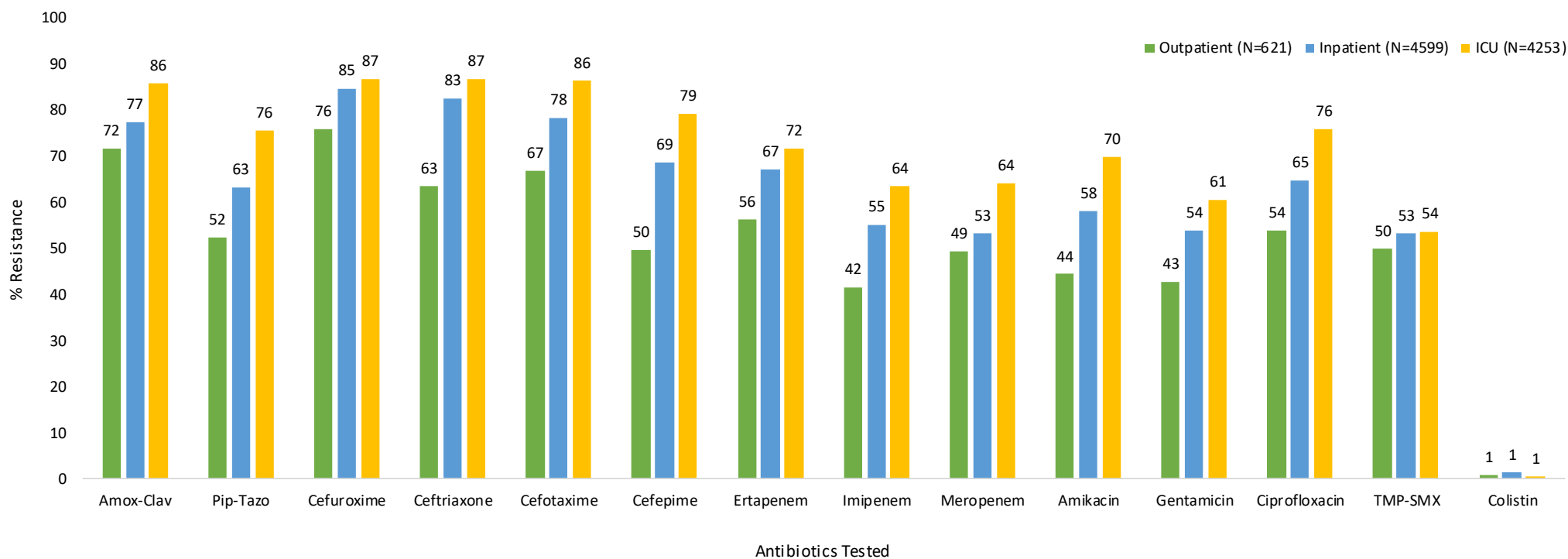
Table 9: Resistance profile of *Klebsiella* species (N=52,597)

	Blood (N=9,478)			PA (N=16,379)			OSBF (N=3,586)			Urine (N=23,154)		
Antibiotic Tested	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI
Amox-Clav	7545	81	80.2-82.0	12600	73	72.0-73.6	2646	71	69.6-73.1	17569	67	66.3-67.7
Pip-Taz	8289	68	67.4-69.4	14085	60	59.2-60.8	3064	58	56.0-59.5	19785	52	51.0-52.4
Cefuroxime	4737	85	84.3-86.4	8761	77	76.0-77.7	1665	81	78.7-82.6	11559	70	69.1-70.8
Ceftriaxone	6385	84	82.6-84.4	9988	75	73.9-75.6	2241	77	75.4-78.9	13243	66	65.5-67.1
Cefotaxime	5073	82	80.6-82.8	10884	75	74.6-76.3	1914	75	72.8-76.7	16478	69	68.5-69.9
Cefepime	7620	72	71.4-73.5	12626	65	64.5-66.1	2814	63	60.7-64.3	15348	58	57.1-58.7
Ertapenem	4399	69	2.068652	7873	53	51.8-54.0	1698	54	51.2-56.0	9925	42	41.3-43.2
Imipenem	7682	58	4.425931	13063	46	44.9-46.7	2729	46	44.2-48.0	16804	37	36.1-37.6
Meropenem	6913	58	2.777376	13097	47	45.9-47.6	2595	48	45.6-49.5	15612	36	35.4-37.0
Amikacin	8563	63	61.7-63.7	14132	52	50.9-52.6	3113	47	45.2-48.8	19241	45	44.0-45.4
Gentamicin	7673	56	3.375473	12502	51	50.0-51.8	2830	47	45.0-48.7	18584	43	42.6-44.0
Ciprofloxacin	7739	69	68.4-70.5	13091	70	68.8-70.4	2781	65	62.9-66.5	17394	64	63.6-65.1
TMP-SMX	7533	53	52.1-54.4	12647	58	56.7-58.4	2845	52.5	50.6-54.3	18925	52	51.4-52.8
Colistin	6132	1.0	0.8-1.3	9203	0.3	0.2-0.4	2101	0.7	0.4-1.1	12818	0.4	0.3-0.5
Doxycycline	X			3864	37	35.8-38.9	1159	37	34.7-40.3	X		
Nitrofurantoin	X			X			X			20870	58	57.1-58.4

*Alert pathogens confirmed at NRL, NCDC only were included in the data

x- Drug bug combination for the specimen type not included in NARS-Net surveillance panel

TMP-SMX – Trimethoprim-Sulfamethoxazole
Amox-Clav – Amoxicillin-Clavulanic acid
Pip-Taz – Piperacillin-Tazobactam



*Data from the emergency department was clubbed with data from inpatient wards; the Location type for 5 isolate is unknown

Amox-Clav – Amoxicillin- Clavulanic acid
PipTaz – Piperacillin-Tazobactam
TMP-SMX – Trimethoprim-Sulfamethoxazole

Figure 13: Resistance profile of *Klebsiella* species isolated from blood (N=9,478) by location type

Salmonella enterica serovar Typhi and Paratyphi

In the current reporting period, data of 1176 *Salmonella enterica* serovar Typhi and Paratyphi isolates was submitted to NCDC. Thirty-one of these isolates were from stool specimens and 1145 isolates were from blood. Among the first line antibiotics, highest resistance was observed to ciprofloxacin (42%). Low resistance to other first-line antibiotics like ampicillin (10%), chloramphenicol (6%) and trimethoprim-sulfamethoxazole (6%) was observed (Table 10). Thirty-eight isolates of *S. enterica* serovar Typhi were found to be resistant to ceftriaxone and sixteen isolates were resistant to azithromycin. Three isolates of ceftriaxone resistant *S. enterica* serovar Paratyphi were observed during 2025. One isolate each of *Salmonella* Typhi and Paratyphi A were found to be resistant to imipenem. All the ceftriaxone resistant, azithromycin resistant and imipenem resistant *S. Typhi* isolates included in this report were confirmed at the AMR-NRL at NCDC.

Table 10: Resistance profile of *Salmonella enterica* Typhi and Paratyphi (N=1145) from blood

Antibiotic tested	<i>S. Typhi</i> (N=1002)			<i>S. Paratyphi</i> A (N=143)		
	Number tested	R (%)	95% CI	Number tested	R (%)	95% CI
Ampicillin	821	10	8.3-12.6	120	8	3.7-14.2
Azithromycin	827	2	1.2-3.3	x		
Ceftriaxone	957	4	2.7-5.2	130	2	0.3-6.0
Chloramphenicol	799	6	4.3-7.7	112	1	0-5.6
Ciprofloxacin	958	42	39.0-45.4	131	27	19.5-35.3
Imipenem	941	0.9	0.4-1.7	136	1	0-4.6
Pefloxacin	640	91	88.0-92.7	87	86	76.8-92.4
TMP-SMX	936	6	4.7-7.9	118	2	0.3-6.6

Alert pathogens were confirmed at NRL, NCDC were included in the data for analysis; TMP-SMX: Trimethoprim Sulfamethoxazole

*R stands for Resistance

Table 11: Resistance profile of *Salmonella enterica* Typhi and Paratyphi (N=31) from stool specimens

Antibiotic tested	<i>S. Typhi</i> (N=14)		<i>S. Paratyphi</i> A (N=17)	
	Number tested	Number Resistant	Number tested	Number Resistant
Ampicillin	13	3	15	3
Azithromycin	11	1		
Ceftriaxone	13	1	15	2
Chloramphenicol	12	8	13	0
Ciprofloxacin	14	1	17	6
Imipenem	13	0	16	1
Pefloxacin	11	11	12	9
TMP-SMX	13	2	15	1

Shigella species

In the current reporting period, data of 120 *Shigella* species isolates from stool specimens was submitted. In these *Shigella* species isolates, resistance was found to be highest to ciprofloxacin (81%) followed by ampicillin (77%) and Trimethoprim/ Sulfamethoxazole (55%). Least resistance was observed to chloramphenicol (3%) (Table 11).

Table 11: Resistance profile of *Shigella* species (N=120) from stool specimens

Antibiotics tested	Number tested	Resistance (%)	95% CI
Ampicillin	110	77	68.1-84.5
Azithromycin	105	31	22.9-41.3
Ceftriaxone	113	45	35.8-54.8
Chloramphenicol	96	3	0.8-9.5
Ciprofloxacin	105	81	71.9-87.7
Trimethoprim/ Sulfamethoxazole	106	55	44.8-64.3

1.2.3.2.2. Non-Fermenting Gram-Negative Bacilli

Among the non-fermenting Gram-Negative bacilli (NF GNB) included in the data submitted during January-December 2025 from the NARS-Net sentinel surveillance sites, *Pseudomonas* species was the most frequently isolated pathogen from 26,637 isolates followed by *Acinetobacter* species (22,711). Among the NF GNB, *Pseudomonas* species was the predominant isolate from inpatients, while *Acinetobacter* species was the predominant isolate from patients in ICU settings (Table 5).

Pseudomonas species

Pseudomonas spp. isolates included in the data during current reporting period were most commonly isolated from pus aspirates (48%), urine (28%), blood (16%), and other sterile body fluid (8%) specimens (Table 4).

Blood culture isolates of *Pseudomonas* species showed highest resistance (37%; CI: 35.9-38.9) to third-generation cephalosporin (ceftazidime) followed by amikacin (29%) 27.6-30.6, ciprofloxacin (27%; CI: 26.0-28.9) and gentamicin (27%; CI: 25.0-28.3) amongst first line agents for treating *Pseudomonas* species infections. Among pus aspirates, ceftazidime and ciprofloxacin showed similar resistance i.e. 42% & 43% respectively. The resistance to ceftazidime and ciprofloxacin in isolates from other sterile body fluids was 36% & 35% respectively. Among the second line antibiotics, resistance to imipenem and aztreonam was less than 40% (Table 12).

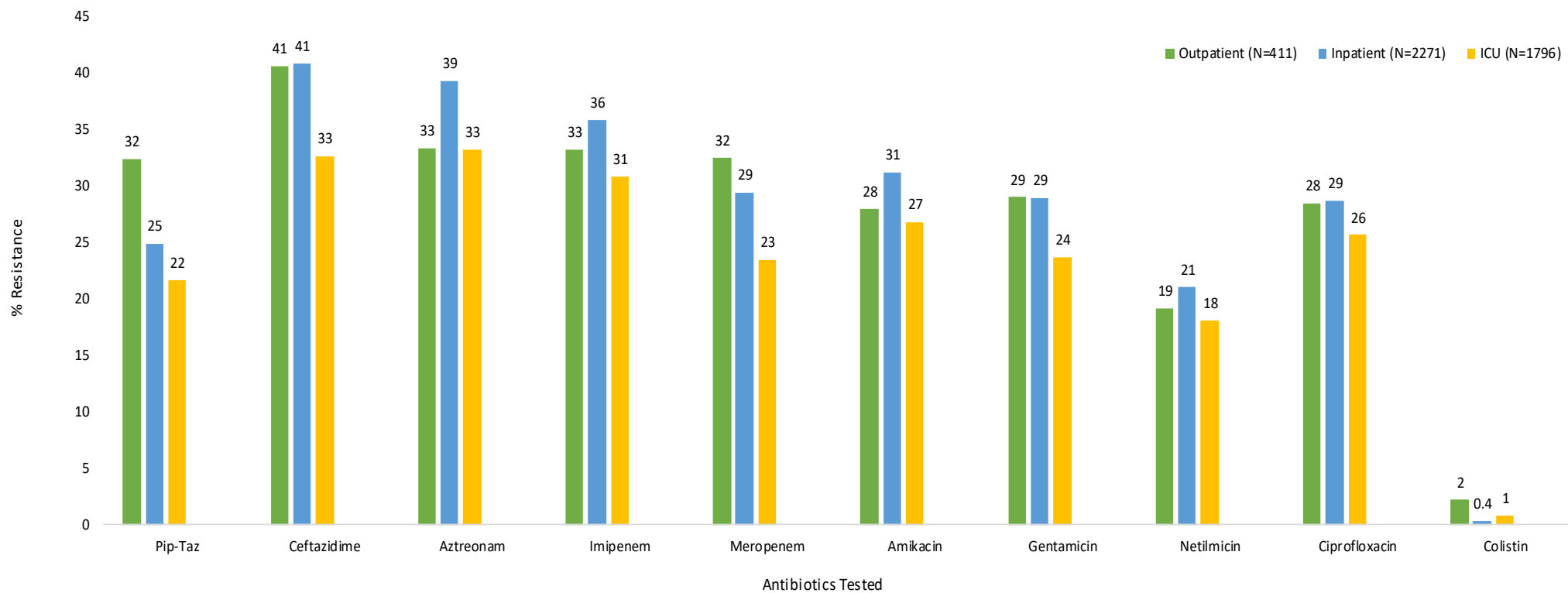
Among urinary isolates, 54% *Pseudomonas* spp. isolates were resistant to ciprofloxacin, 53% to ceftazidime, 45% to gentamicin, 40% to aztreonam and 33% to piperacillin-tazobactam. (Table 12). For the reserve group of antibiotic, namely colistin, twenty isolates from blood cultures (0.7%), four from pus aspirate (0.1%) and four from urine (0.1%) were found to be resistant (Table 12).

Isolates of *Pseudomonas* spp. from blood cultures of patients in Intensive care units, Inpatient and outpatient departments showed highest resistance to ceftazidime and aztreonam among the tested antibiotics (Fig. 14). Among blood isolates from ICU patients, low resistance was observed to netilmicin (18%), piperacillin-tazobactam (22%) and meropenem (23%) (Fig. 14).

Table 12: Resistance profile of *Pseudomonas* species (N=26,637)

Antibiotic Tested	Blood (N=4,478)			Pus Aspirate (N=12,739)			OSBF (N=2,369)			Urine (N=7,051)		
	Number Tested	(%R)	95 % CI	Number Tested	(%R)	95 % CI	Number Tested	(%R)	95 % CI	Number Tested	(%R)	95 % CI
Amikacin	3572	29	27.6-30.6	9584	33	31.9-33.8	1805	24	21.8-25.8	6091	42	40.4-42.9
Aztreonam	2960	36	34.4-37.9	8322	27	25.9-27.8	1584	29	26.5-31.0	4862	39	37.2-39.9
Ceftazidime	3906	37	35.9-38.9	11146	42	40.9-42.7	2098	36	34.1-38.2	6216	53	51.8-54.3
Ciprofloxacin	3757	27	26.0-28.9	10328	43	41.7-43.6	1947	35	32.4-36.7	5628	54	53.0-55.6
Colistin	2886	0.7	0.5-1.1	7200	0.1	0.1-0.3	1537	0.5	0.2-1.1	4246	0.1	0-0.2
Gentamicin	2866	27	25.0-28.3	7406	36	34.7-36.9	1457	24	22.3-26.7	4488	45	43.5-46.4
Imipenem	3850	33	32.0-35.0	10743	28	27.4-29.2	1996	31	28.8-32.8	5787	43	41.3-43.9
Meropenem	3588	27	25.7-28.6	10455	27	25.7-27.4	1582	30	28.0-32.6	4776	40	38.8-41.6
Netilmicin	2224	19	17.9-21.2	6217	30	28.9-31.2	1151	22	19.6-24.4	3686	41	39.4-42.6
Piperacillin/ Tazobactam	4017	24	23.0-25.6	11310	27	26.0-27.6	2082	23	21.5-25.1	6355	33	31.6-33.9

Alert pathogens confirmed at NRL, NCDC only were included in the data



*Data from the emergency department was clubbed with data from inpatient wards

PipTaz – Piperacillin-Tazobactam

Figure 14: Resistance profile of *Pseudomonas* species isolated from blood (N=4,478) by location type

Acinetobacter spp.

Data of a total of 22,711 *Acinetobacter* species isolates were submitted by network sites during this reporting period of Jan – Dec 2025. Among all specimen types included under the programme, *Acinetobacter* species was most commonly isolated from blood (39%) followed by pus aspirates (30%), urine (18%) and other sterile body fluid specimens (12%) (Table 4).

High resistance was observed to all the tested antibiotics including Minocycline in isolates from blood, pus specimens and other sterile body fluids in ICU patients (Fig. 15). Blood isolates showed the highest resistance to ceftazidime (75%; CI: 74.2-76.2) followed by Imipenem (70%; CI: 69.0-71.1). Alarming high level of resistance to second line antimicrobial agents viz. imipenem (70%) and meropenem (67%) was observed among the blood isolates (Table 13). Sixty-nine percent of isolates from pus specimens and 62% from other sterile body fluids were resistant to imipenem. Twenty-five isolates from blood, two from OSBF, 7 from pus aspirates and 4 isolates of urine were resistant to colistin (Table 13).

Table 13: Resistance profile of *Acinetobacter* species (N=22,711)

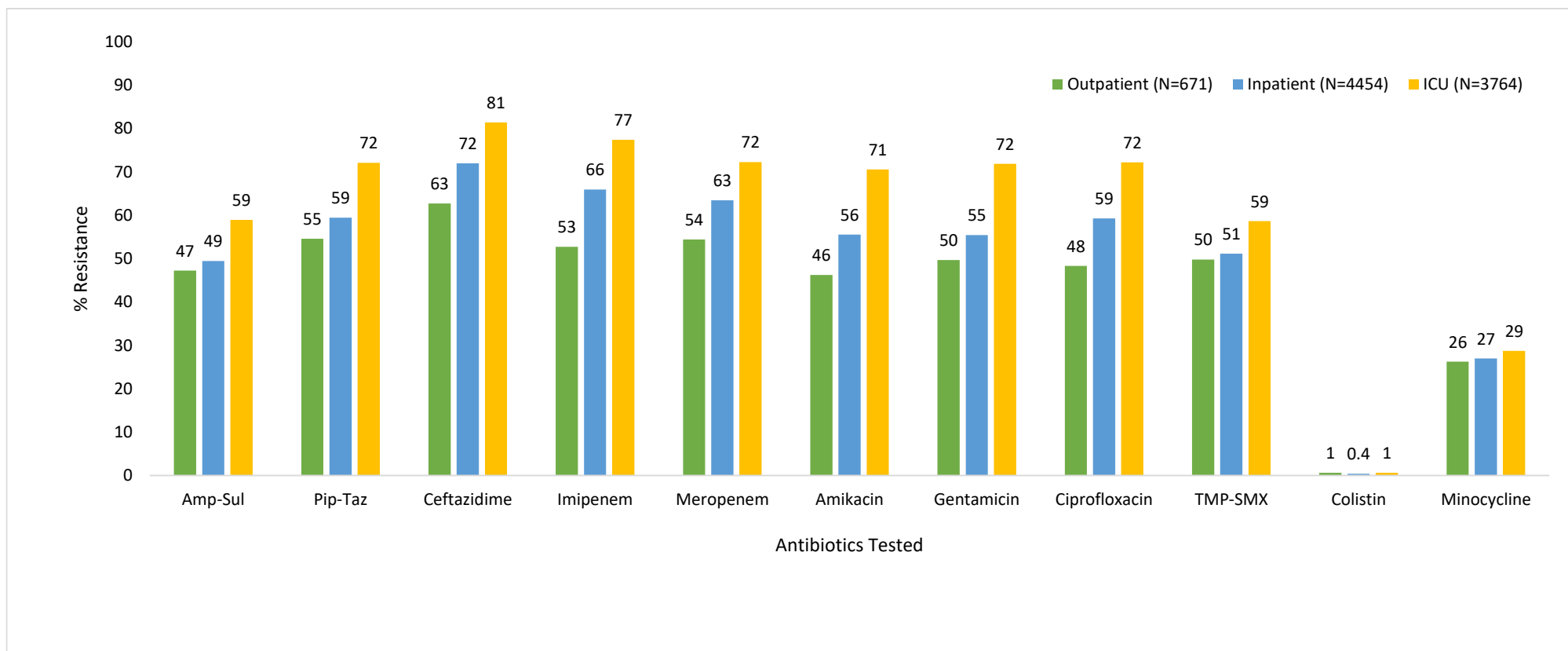
	Blood (N=8,890)			OSBF (N=2,972)			PA (N=6,804)			Urine (N=4,045)		
Antibiotic Tested	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI	Number Tested	(%R)	95% CI
Amp-Sul	4144	54	52.2-55.2	1532	45	42.6-47.6	3713	60	58.7-61.9	1742	38	36.1-40.7
Pip-Taz	7664	65	63.6-65.7	2610	58	56.2-60.1	5874	69	67.8-70.2	3265	43	41.3-44.7
Ceftazidime	7044	75	74.2-76.2	2335	69	66.8-70.6	5037	77	76.1-78.5	2983	62	60.4-63.9
Imipenem	7240	70	69.0-71.1	2076	62	59.9-64.1	5392	69	67.8-70.3	2873	43	41.6-45.2
Meropenem	6159	67	65.4-67.8	2264	59	56.8-60.9	5407	66	64.9-67.5	2223	41	38.5-42.7
Amikacin	7885	61	60.3-62.5	2648	53	50.8-54.6	5745	67	66.2-68.7	2987	42	40.7-44.2
Gentamicin	7061	62	61.3-63.5	2548	53	51.2-55.1	5463	66	64.8-67.4	3096	42	40.2-43.7
Ciprofloxacin	7029	64	63.1-65.3	2097	59	57.2-61.4	5331	73	71.4-73.8	2983	52	50.5-54.1
TMP-SMX	6621	54	53.1-55.6	2215	55	52.8-56.9	4802	70	68.6-71.2	2949	45	43.5-47.1
Colistin	5071	0.5	0.3-0.8	1855	0.1	0-0.4	3647	0.2	0.1-0.4	1764	0.2	0-0.5
Minocycline	5346	28	26.5-29.0	1879	23	20.9-24.7	3842	32	30.4-33.3	1896	24	22.2-26.1
Tetracycline	X			x			x			1549	42	39.3-44.3

*Alert pathogens confirmed at NRL, NCDC only were included in the data

TMP-SMX: Trimethoprim-Sulfamethoxazole

Amp-Sul: Ampicillin-Sulbactam

Pip-Taz: Piperacillin-Tazobactam



* Data from the emergency department was clubbed with data from inpatient wards; the Location type for 1 isolate is unknown.

Figure 15: Resistance profile of *Acinetobacter* species isolated from blood (N=8,890) by location type

1.2.3.2.3. *Vibrionaceae*

Vibrio cholerae

In the current reporting period (Jan – Dec 2025), data of 106 isolates of *Vibrio cholerae* confirmed at AMR-NRL has been analyzed. The highest resistance was observed to trimethoprim/sulfamethoxazole and ampicillin i.e. 18% for both while no resistance to chloramphenicol. All isolates were not tested to all surveillance panel antibiotics. (Table 14)

Table 14: Resistance profile of *Vibrio cholerae* (N=106)

Antibiotic Tested	Number Tested	Resistant (%)	95% CI
Ampicillin	95	18	11.1-27.4
Trimethoprim/ Sulfamethoxazole	103	18	11.7-27.6
Azithromycin	86	5	1.5-12.1
Chloramphenicol	91	0	0.0-5.0
Doxycycline	88	1	0.1-7.1
Tetracycline	96	2	0.4-8.0

1.2.4. AMR Trends

AMR Surveillance Data from NARSNET sites has been analyzed over last 8 years from 2017 to 2025. During 2020, data reported was small in volume due to the COVID pandemic. Trend analysis of MRSA isolated from blood showed slight increase in MRSA proportion from 59% (2021) to 61% (2025) (Fig. 16). This increase in proportion of MRSA may be attributed to the increase in data volume as submitted by the network sites. Trend analysis of VRE isolated from blood showed continuous increase from 2021 (11%) to 2025 (30%) (Fig. 18). Trend analysis for ESBL producing *E. coli* isolated from blood showed an increase over the last 4 years from 2021 (76%) to 2024 (81%), in 2025 the trend is similar to the previous year (80%) (Fig. 19). The trends observed for ESBL producing *Klebsiella* spp. from blood specimens showed consistently high level of resistance ranging from 81% (2021) to 84% (2024) and in 2025 similar resistance level was observed (83%) (Fig. 19). Four-year trend analysis of carbapenemase producing *Klebsiella* species has shown consistent increase (Fig 21, 22). While for other organisms like *E. coli*,

Pseudomonas spp., and *Acinetobacter* spp., the resistance to carbapenems has slightly decreased which may be due to increase in the volume of data received over the years.

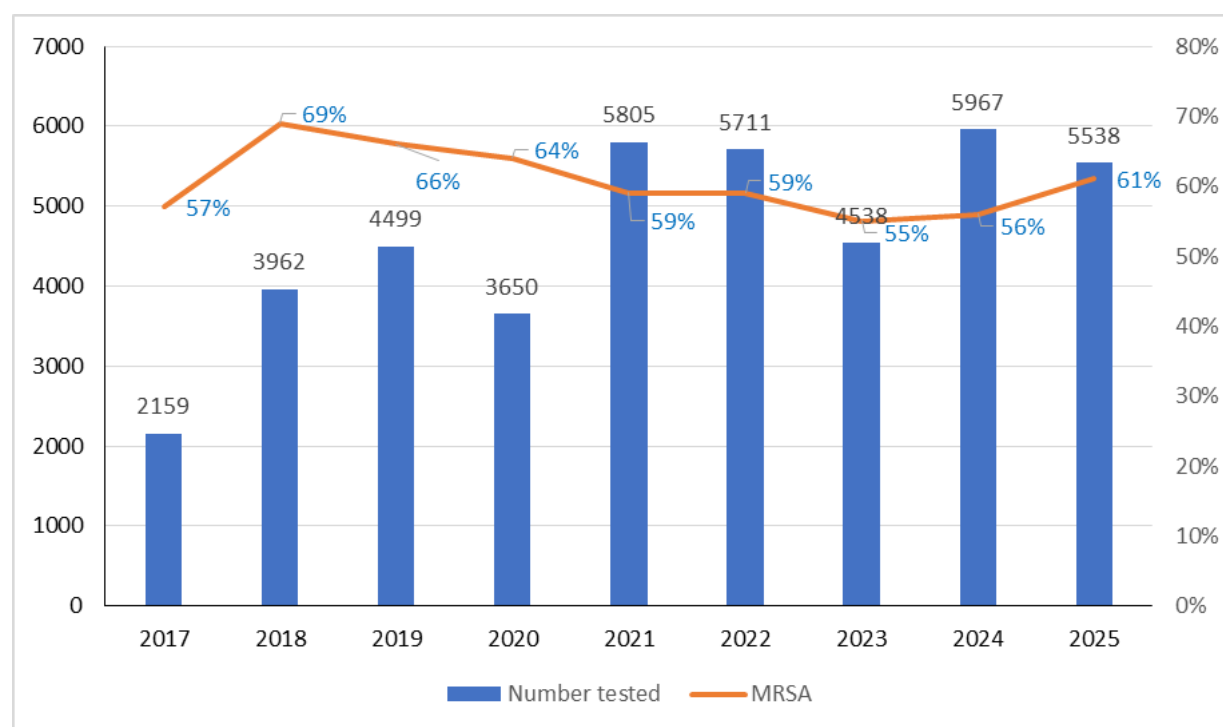
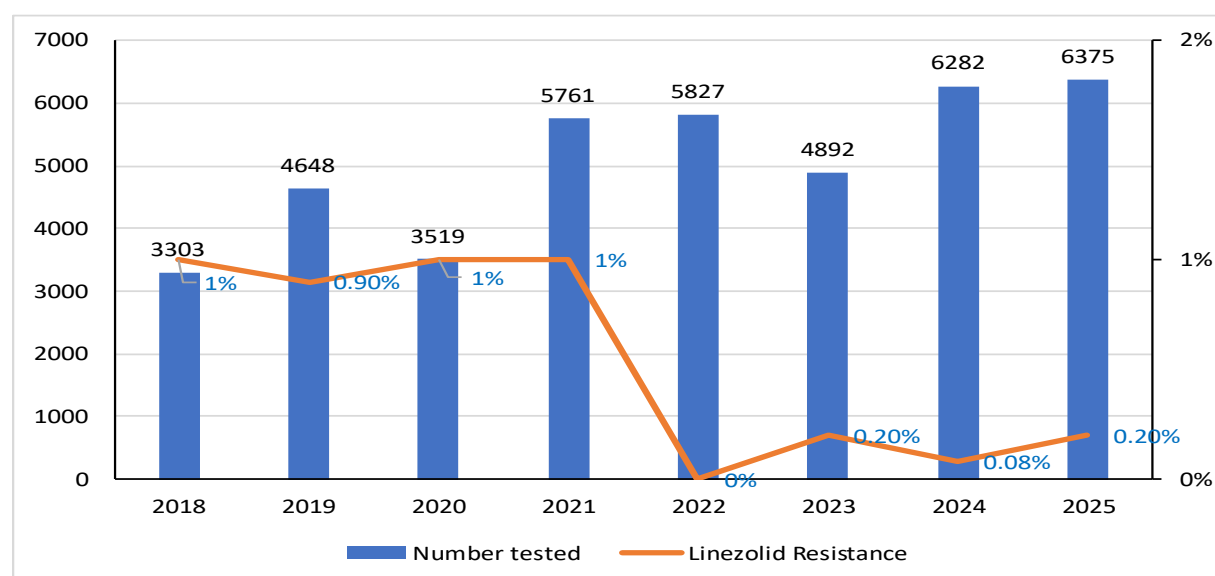
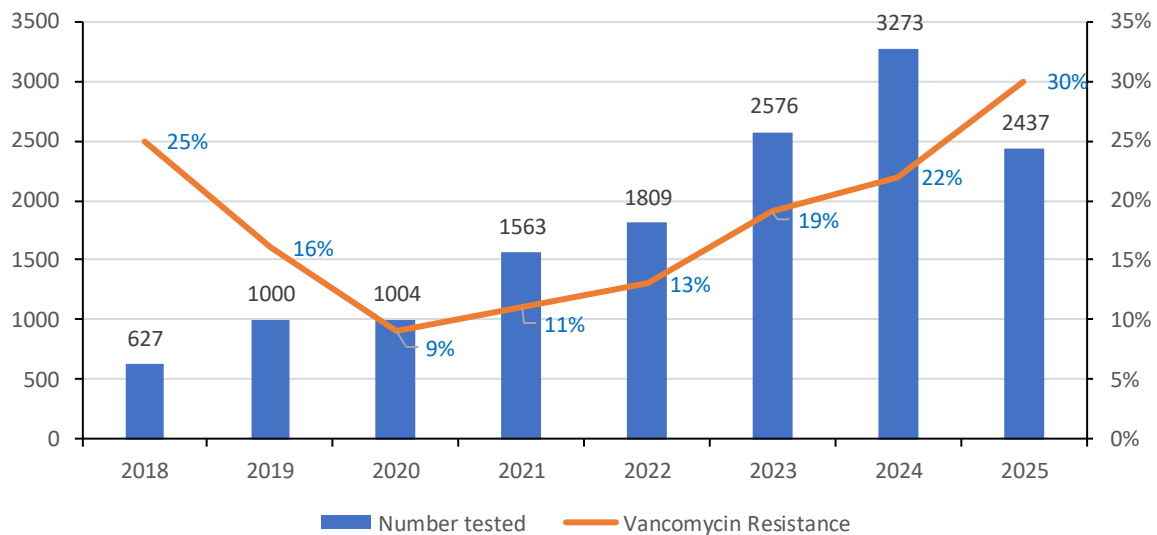


Figure 16: Trends of Methicillin-resistant *S. aureus* (MRSA) isolated from blood (2017-2025)



**Alert pathogens confirmed at NRL, NCDC only were included in the data; CLSI breakpoint for 2023-25 were considered for analysis of linezolid resistance in S.aureus*

Figure 17: Trends of Linezolid resistant *Staphylococcus aureus* isolated from blood (2018 to 2025)



*Alert pathogens confirmed at NRL, NCDC only were included in the data

Figure 18: Trends of Vancomycin resistant *Enterococcus* sp. (VRE) isolated from blood (2018-2025)

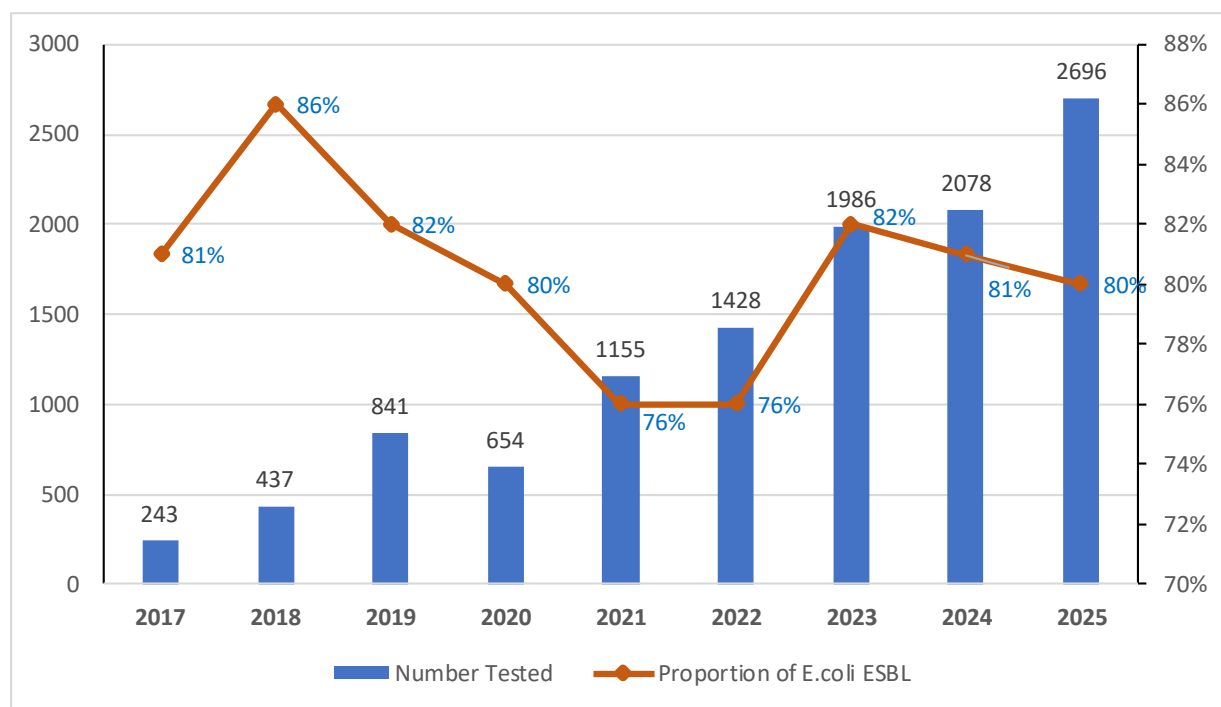


Figure 19: Trends of Extended spectrum beta-lactamase (ESBL) producing *E. coli* isolated from blood (2017-2025)

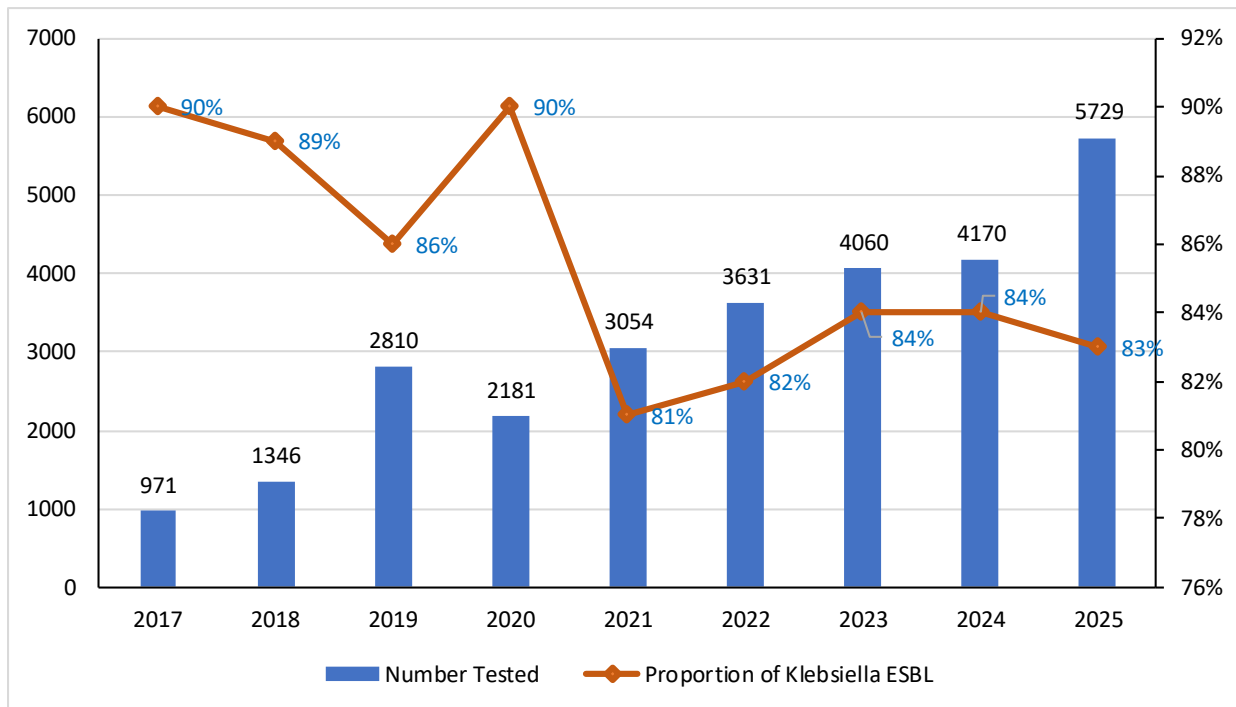


Figure 20: Trends of Extended-Spectrum Beta-Lactamase (ESBL) producing *Klebsiella* spp. in blood (2017-2025)

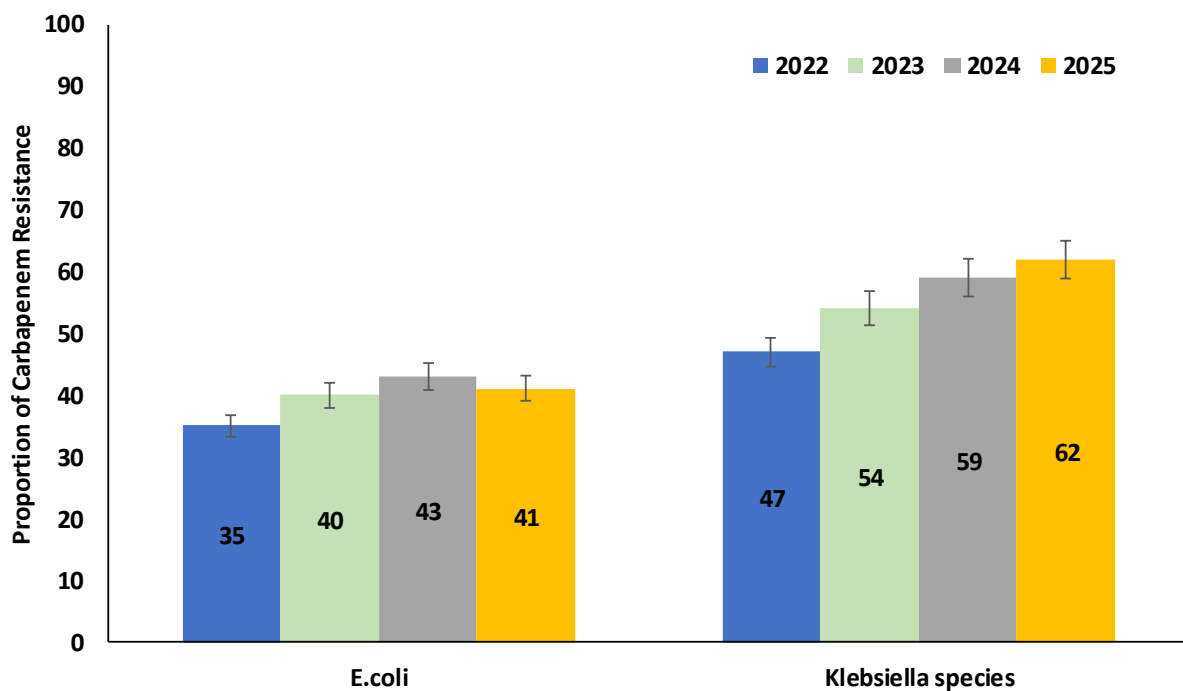


Figure 21: Four-year trend analysis (2022-2025) of Carbapenemase producing *E. coli* and *Klebsiella* species isolated from blood

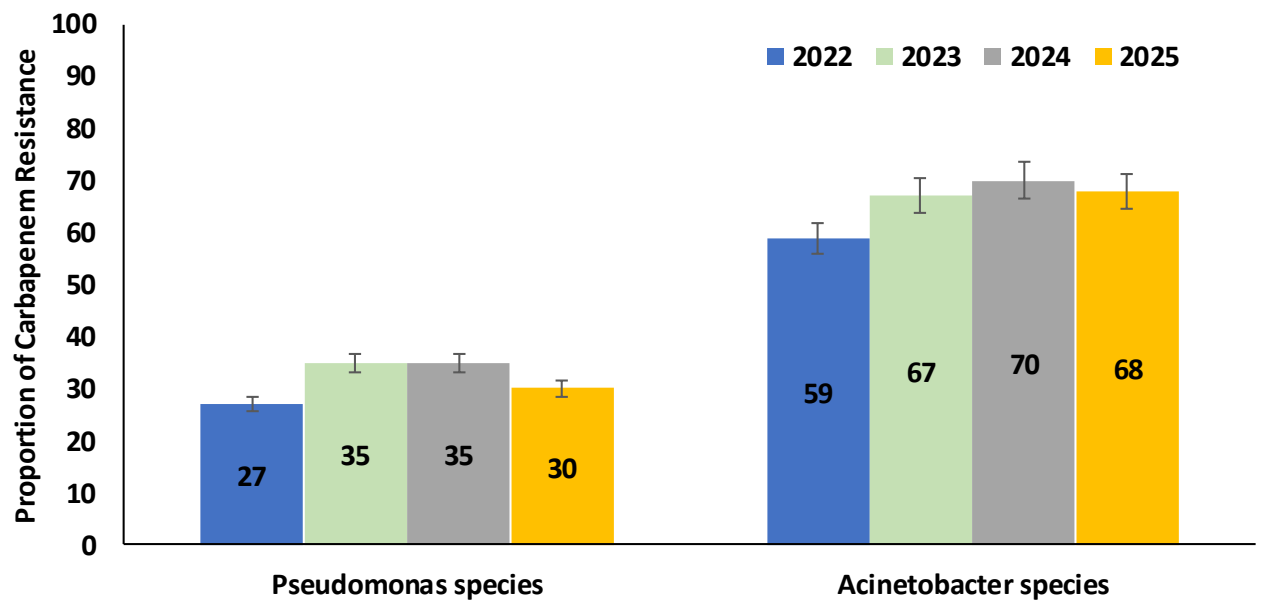


Figure 22: Four-year trend analysis (2022-2025) of Carbapenemase producing *Pseudomonas* species and *Acinetobacter* species isolated from blood

1.2.5. Molecular Detection and Characterization of Priority Bacterial Pathogens

Under the National Programme on Antimicrobial Resistance (AMR) Containment, the Antimicrobial Resistance National Reference Laboratory (AMR-NRL) for bacterial pathogens established at the Centre for Bacterial Diseases and Drug Resistance Research (CBDDR), National Centre for Disease Control (NCDC), Delhi, received a total of 2,197 carbapenem-resistant Gram-negative bacterial isolates and 170 vancomycin-resistant *Enterococcus* spp. NARS-Net network sites for confirmation as a part of the External Quality Assurance Scheme (EQAS) and as emerging AMR alerts.

Among the 2,197 carbapenem-resistant Gram-negative isolates, 509 (23%) were *Escherichia coli*, 744 (34%) were *Klebsiella* spp., 527 (24%) were *Acinetobacter* spp., and 417 (19%) were *Pseudomonas* spp. All isolates were confirmed at the AMR-NRL, CBDDR, NCDC using standard biochemical identification methods and phenotypic antimicrobial susceptibility testing in accordance with established laboratory protocols.

For molecular characterization of carbapenem resistance, 522 representative isolates were randomly selected from different geographical regions of India, including North, South, East, West, Central, and North-East India, to ensure regional representation. Conventional polymerase chain reaction (PCR) was performed to detect the New Delhi metallo- β -lactamase (NDM) and Oxacillinase-48 (OXA-48) carbapenemase genes. The selected isolates comprised 136 *Escherichia coli*, 152 *Klebsiella* spp., 136 *Acinetobacter* spp., and 98 *Pseudomonas* spp.

For molecular characterization of vancomycin resistance, all 170 vancomycin-resistant *Enterococcus* spp. received at the AMR-NRL, CBDDR, NCDC were analysed by conventional PCR for the detection of the Vancomycin Resistance Gene Cluster A (vanA) and Vancomycin Resistance Gene Cluster B (vanB) genes.

1.2.5.1 Molecular characterization of Carbapenemase-Encoding Genes (*bla*NDM and *bla*OXA-48) in Gram-Negative Bacterial Isolates

Among the 152 carbapenem-resistant *Klebsiella* spp. (CRK) isolates tested for NDM and OXA-48 carbapenemase genes by PCR, 75 (49.34%) were positive for the NDM gene, 8 (5.26%) carried the OXA-48 gene, and 64 (42.10%) harboured both NDM and OXA-48 genes. These findings indicate a high prevalence of NDM-mediated carbapenem resistance among the *Klebsiella* spp. isolates and a substantial proportion of isolates carrying both NDM and OXA-48 genes.

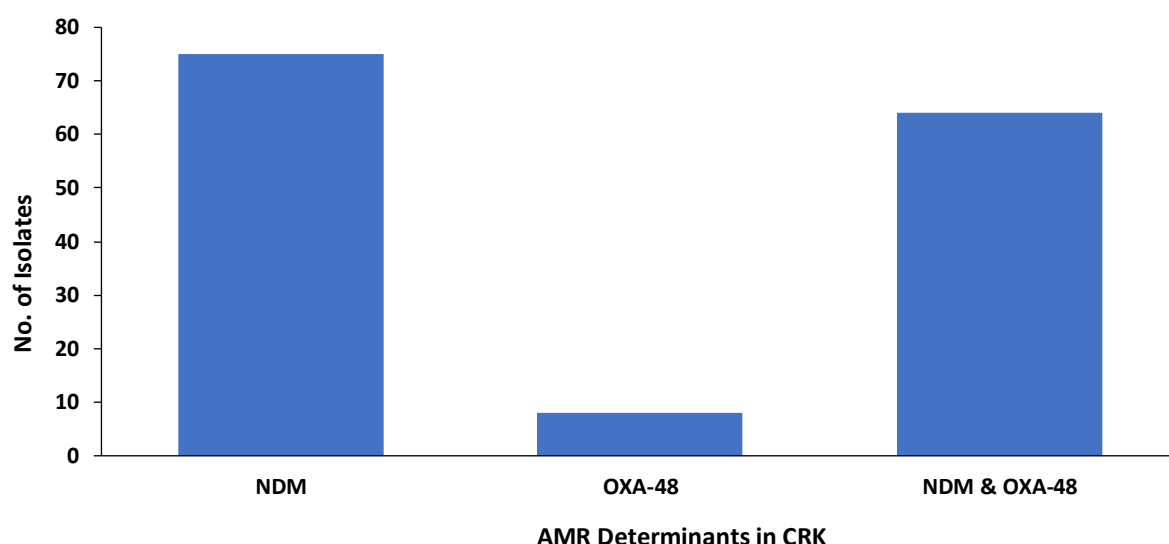


Figure 23: Distribution of NDA & OXA-48 gene in *Klebsiella* species

Out of 136 carbapenem-resistant *Escherichia coli* (CRE) isolates tested for presence of NDM and OXA-48 carbapenemase genes by PCR, 88 (64.70%) were positive for the NDM gene, 3 (2.20%) carried the OXA-48 gene, and 39 (28.67%) harboured both NDM and OXA-48 genes. These findings indicate that NDM is the predominant carbapenem resistance gene among the *E. coli* isolates, either alone or in combination with OXA-48.

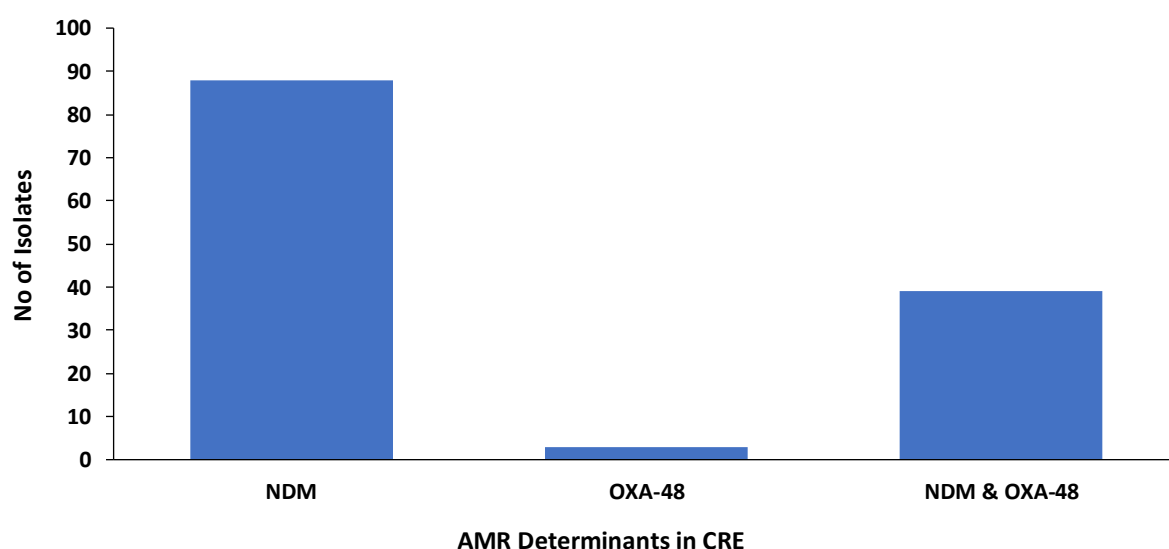


Figure 24: Distribution of NDA & OXA-48 gene in *E. coli*

PCR to detect the presence of NDM and OXA-48 carbapenemase genes performed on 136 carbapenem-resistant *Acinetobacter* spp. (CRA) isolates, showed 92 isolates (67.64%) having the NDM gene, 5 (3.67%) carried the OXA-48 gene, and 14 (10.29%) harboured both NDM and OXA-48 genes. These findings indicate that NDM is the predominant carbapenem resistance gene in the *Acinetobacter* spp. isolates.

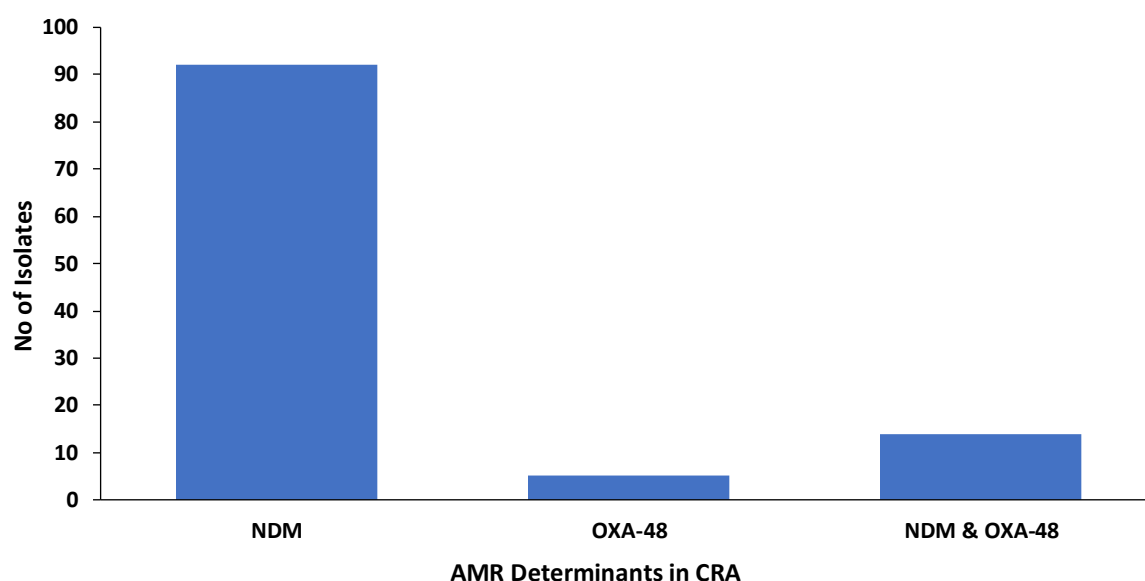


Figure 25: Distribution of NDA & OXA-48 gene in *Acinetobacter* species

PCR was also performed on 98 carbapenem-resistant *Pseudomonas* species (CRP) isolates to detect the presence of NDM and OXA-48 carbapenemase genes. Among the 98 isolates tested, 67 (68.36%) were positive for the NDM gene, 3 (3.06%) carried the OXA-48 gene, and 8 (8.16%) harboured both NDM and OXA-48 genes. These findings indicate that NDM is the predominant carbapenem resistance gene among the *Pseudomonas* species isolates.

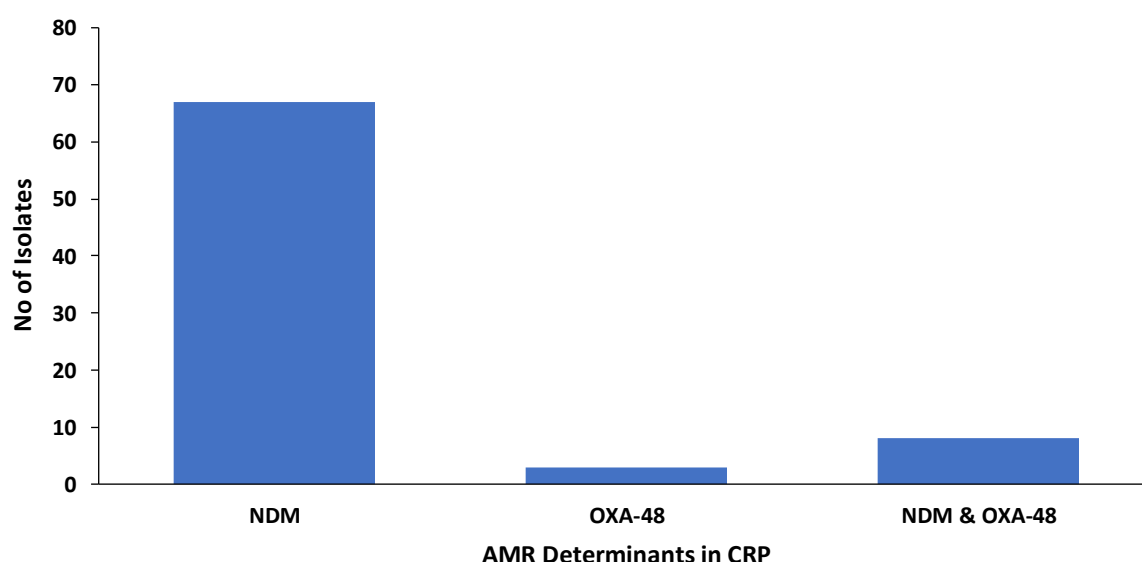


Figure 26: Distribution of NDA & OXA-48 gene in *Pseudomonas* species

1.2.5.2 Molecular Characterization of Vancomycin Resistant Genes (*vanA* and *vanB*) in *Enterococcus* species

A total of 157 vancomycin-resistant *Enterococcus* spp. isolates were screened for the presence of *vanA* and *vanB* genes by PCR. Among these, 119 isolates (75.79%) were positive for

the *vanA* gene, while none of the isolates carried the *vanB* gene. The results indicate that the *vanA* gene is the predominant mechanism of vancomycin resistance among the *Enterococcus* species isolates.

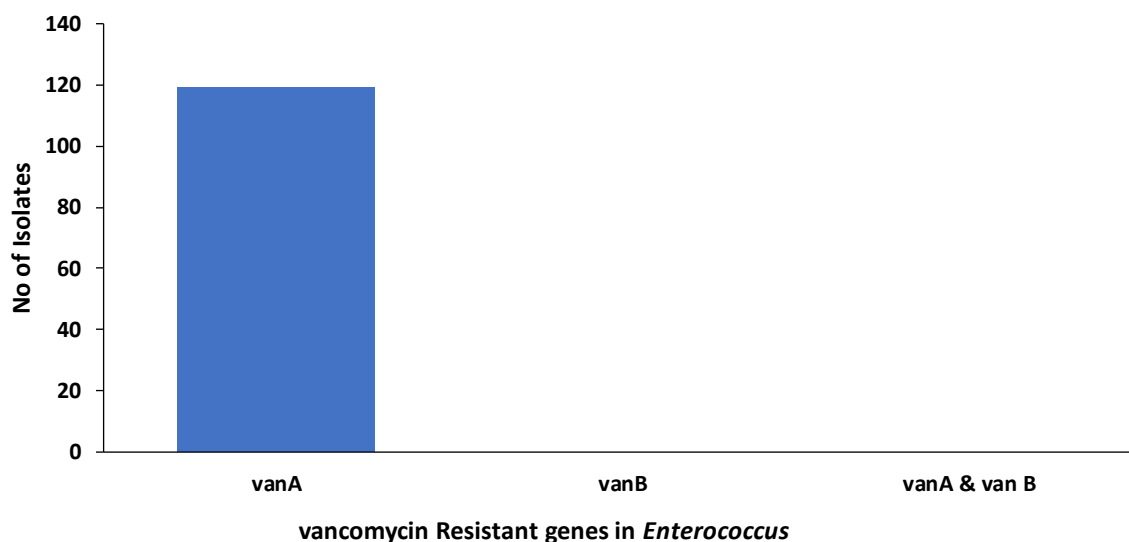


Figure 27: Distribution of *vanA* & *vanB* genes in *Enterococcus* species

1.2.6. Discussion

This report is the ninth annual report of the National AMR Surveillance Network (NARS-Net). This report includes 2,17,513 isolates AST data submitted by 62 sites. Like the previous year's findings, one-third of the pathogens in the AMR Surveillance data for 2025 were *Escherichia coli* (31%). Among the urinary isolates, *E. coli* (44.5%) was the most commonly isolated priority pathogen, and urine was the most common specimen type (45%). As consistent with previous reports, *S. aureus* was most commonly isolated from pus aspirates (67%). *Klebsiella* species was most commonly isolated pathogen from urine (44%) and pus specimens (31%).

Quality of AMR surveillance data submitted by the NARSNET sites has been ensured by continuous hands-on laboratory and data management trainings, site support visits, monthly data quality monitoring and feedback calls, and annual review meetings. EQAS program and confirmation of AMR Alert isolates at AMR-NRL, CBDDR, NCDC also played an essential role in improving data quality of NARS-Net sites. Moreover, many of the sites are strictly performing internal quality control of media and antibiotic discs as per recommendations under the programme.

Compared to the previous year, the number of sites submitting data has increased from 54 to 62 sites situated in 34 states/UTs. Quality of colistin susceptibility testing has improved over the years, as more than 50 sentinel sites have standardized and are performing colistin

agar dilution and colistin broth microdilution (BMD) tests. The wet lab trainings on agar dilution and broth microdilution are conducted on an annual basis as per need for NARS-Net sites.

The AMR surveillance data in the current report like previous years continues to have the limitation of samples for culture and sensitivity being referred only for patients requiring admission and often not responding to the first line of antibiotics. This limitation has the potential to skew the AMR trends to higher side than the actual trends.

Trend analysis over last 5 years for ESBL producing *E. coli* from blood has shown high prevalence over the years (Fig. 19). This may be attributed to higher usage of third generation cephalosporins for all the Gram-negative priority surveillance pathogens. High percentage of resistance to third-generation cephalosporins and carbapenems in *E. coli* and *Klebsiella* species and *Acinetobacter* species in India is a matter of concern. These findings highlight serious challenges in treatment options for patients with infections caused by these priority bacterial pathogens.

There has been a gradual increase in the proportion of Vancomycin-resistant *Enterococcus* (VRE) from blood specimens (from 11% in 2021 to 30% in 2025). (Fig. 18) Linezolid resistance (2.6%) in blood isolates of *Enterococcus* species is slightly higher than that reported last year. No Vancomycin-intermediate *Staphylococcus aureus* (VISA) or Vancomycin-resistant *Staphylococcus aureus* (VRSA) has been reported over last 9 years. Trend analysis of MRSA isolated from blood over the past 5 years shows a gradual increase to 61% in 2025. (Fig 16). Among MRSA from blood specimens, highest resistance was observed to erythromycin (75%) followed by ciprofloxacin (73%) and Clindamycin (48%) (Fig 8). The data of *S. aureus* continues to be affected by the challenge of accurate identification of *S. aureus* using phenotypic methods like hemolysis on sheep blood agar, slide coagulase and Mannitol Salt Agar.

An increase in *Salmonella* Typhi isolates has been observed over the years with increasing resistance to the first line antibiotics namely ceftriaxone - 1.4% in 2024 to 4% in 2025. Resistance to azithromycin has increased from 0.3 % in 2024 to 2% in 2025 and resistance to Imipenem has increased from 0 (2024) to 0.9 % in 2025. Regarding sensitivity of *Salmonella* Typhi to fluoro-quinolones, the discrepant rates of resistance against Ciprofloxacin and Pefloxacin in *Salmonella* Typhi isolates maybe explained by the differing baseline affinities of ciprofloxacin and pefloxacin to the targeted enzymes DNA Gyrase (gyr A and gyr B subunits). Ser 83 and Asp 87 mutations in gyr A are most commonly seen, which leads to retained binding in Ciprofloxacin while binding fails in Pefloxacin. Moreover, in comparison to Pefloxacin, ciprofloxacin remains partially active against single mutation strains producing large zones of

inhibition on disk diffusion and the strains are observed to be susceptible. Pefloxacin captures both chromosomal and plasmid mediated resistance.⁶

The AMR surveillance data presented in this report provides critical insights into resistance trends and highlights areas needing focused attention. Strengthening surveillance systems to generate quality AMR surveillance data, local use of data for generating antibiograms to guide judicious use of antibiotics, strengthening infection prevention and control in the health care facilities and the community to reduce infections and improving awareness to promote behavior change towards judicious use of antibiotics are essential pillars for AMR containment in the country. The findings in this report reinforce the value of strengthening surveillance system and a unified national AMR network. Continued investment in laboratory capacity, data quality, and site-level support is essential to build and maintain a resilient national response to antimicrobial resistance.

⁶ Sharma P, Dahiya S, Kumari B, Balaji V, Sood S, Das BK, Kapil A. Pefloxacin as a surrogate marker for quinolone susceptibility in *Salmonella enterica* serovars Typhi & Paratyphi A in India. *Indian J Med Res.* 2017 May;145(5):687-692. doi: 10.4103/ijmr.IJMR_494_16. PMID: 28948961; PMCID: PMC5644305.

2. Fungal AMR Surveillance

2.1 Introduction

The National Reference Laboratory (NRL) for Antimicrobial resistance (AMR) in fungal pathogens is established at Vallabhbhai Patel Chest Institute, New Delhi under the National Programme on AMR Containment since September, 2021. As per the Global AMR Surveillance System (GLASS) requirements, for surveillance of AMR in fungal pathogens, currently one fungal pathogen genus that is *Candida* spp. from blood specimens is included. The National AMR Surveillance network (NARSNet) sites under National Programme on AMR Containment participating in the antifungal surveillance have been given wet-lab trainings on fungal isolation, identification and antifungal sensitivity testing to build their capacity.

2.2 National Reference Laboratory for AMR in Fungal Pathogens

2.2.1 Participating laboratories and *Candida* strains processed at NRL for AMR surveillance

The NRL at VPCI confirmed a total of 997 *Candida* spp. isolates from blood cultures from 39 NARSNet sites for phenotypic and molecular characterization and generation of antifungal susceptibility profile under “National Programme on AMR Containment” from January 2025-December 2025. List of NARS-Net sites participating for AMR surveillance in fungal pathogens is at Annexure 1.

2.2.2 Methodology

Phenotypic characterization of *Candida* isolates: All the stains received at NRL were screened on differential selective media CHROMagar™ *Candida* Medium.

Molecular characterization: All strains were identified by Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry (**MALDI-TOF**, Bruker Daltonics, Bremen, Germany) and by sequencing of internal transcribed spacer region (**ITS**) of ribosomal deoxyribonucleic acid (DNA) and of the **D1/D2** domain of the large subunit ribosomal DNA in case low or no identification was obtained by MALDI.

Antifungal susceptibility testing (AFST)

Broth Microdilution Method: AFST was performed on 997 isolates using the Clinical and Laboratory Standards Institute (CLSI) broth microdilution method (BMD) M27-A4, for antifungals amphotericin B, itraconazole, voriconazole, posaconazole, 5-flucytosine, caspofungin, micafungin and anidulafungin. All antifungals were dissolved in dimethyl sulfoxide (DMSO). Roswell Park Memorial Institute medium (RPMI) 1640 medium with glutamine without

bicarbonate (Sigma) buffered to pH 7 with 0.165 M 3-(N-morpholino) propane sulfonic acid (MOPS; Sigma) was used. Drug- and yeast-free controls were included, and microtiter plates were incubated at 35°C and read visually after 24 h. The minimum inhibitory concentration (MIC) endpoints for all the drugs except amphotericin B were defined as the lowest drug concentration that caused a prominent decrease in growth (50%) in relation to the controls and for amphotericin B, the MIC was defined as the lowest concentration at which there was 100% inhibition of growth compared with the drug-free control wells.

Candida krusei ATCC 6258 and *Candida parapsilosis* ATCC 22019 were included as quality control strains in accordance with CLSI recommendations.

2.2.3 Details of 997 yeasts characterized at NRL during 2025

Most common *Candida* species isolated from bloodstream infection were *Candida tropicalis* (n=443, 45%) followed by *Candida albicans* (n=128, 13%), *Candida parapsilosis* (n=94, 10%), *Pichia kudriavzevii* (n=81, 8%), *Candida auris* (n=43, 4%), *Wickerhamomyces anomalus* (n=43, 4%) and *Cyberlindnera jadinii* (n=32, 3%). Non-albicans *Candida* (NAC) species dominate the distribution, highlighting the epidemiological shift from *C. albicans*.

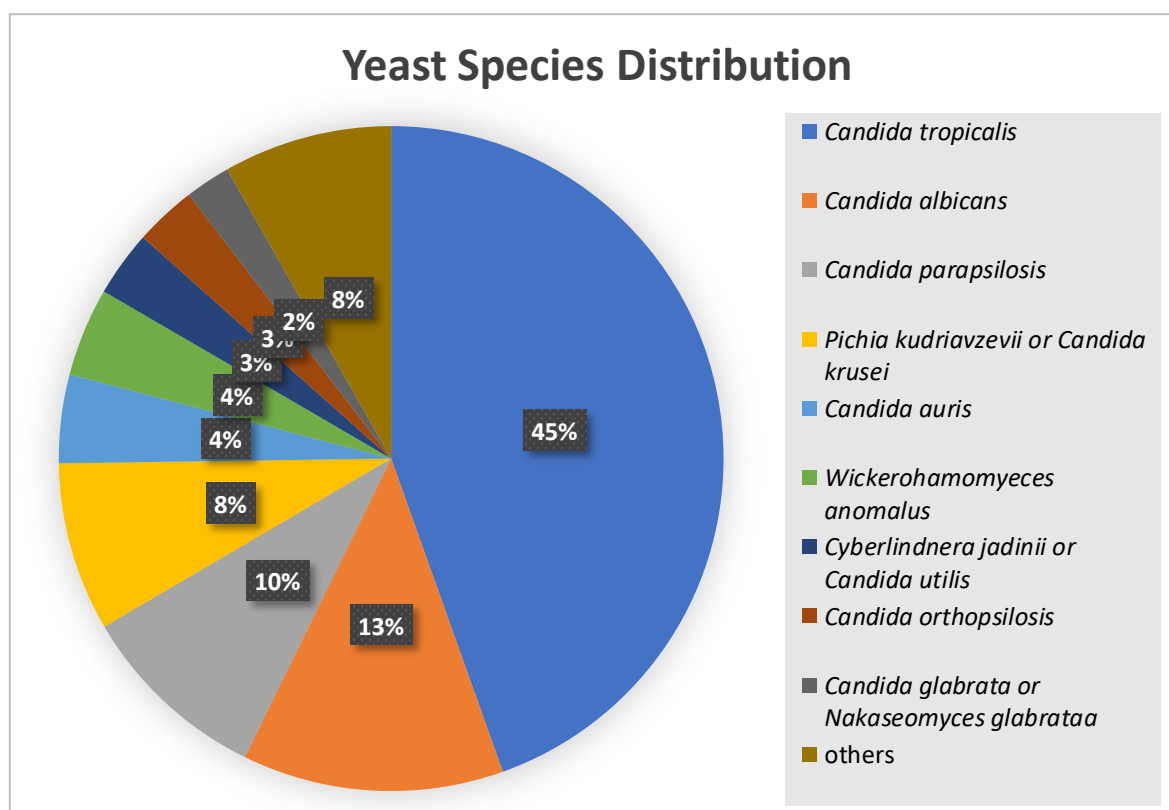


Figure 28: Pie chart illustrating the percentage distribution of various yeast species identified by MALDI-TOF MS and ITS sequencing at NRL, VPCI

Table 15: Distribution of other (8%) yeast species identified at the NRL

S. No.	Other Yeast Species Identified	Number of Isolates
1	<i>Diutina rugosa</i>	16
2	<i>Cyberlindnera fabianii</i>	15
3	<i>Kodamaea ohmeri</i>	14
4	<i>Candida viswanathii</i>	8
5	<i>Meyerozyma guilliermondii</i>	8
6	<i>Candida metapsilosis</i>	7
7	<i>Clavispora lusitaniae</i>	6
8	<i>Candida duobushaemulonii</i>	3
9	<i>Trichosporon asahii</i>	3
10	<i>Candida blankii</i>	1

2.2.3.1 Age and Gender distribution of isolates

Overall, out of all the patients from whose blood culture specimens yeast strains were isolated, 53% were male and 47% were female. Strikingly, neonates and infants (<1 year) account for the majority of the isolates (50%), (Figure 3). Further, young adults (18-34 years) accounted for 9% isolates and 11% were from middle aged (35-54 years) adults. The elderly group (65 years and above) represented 5% of isolates. (Fig 30)

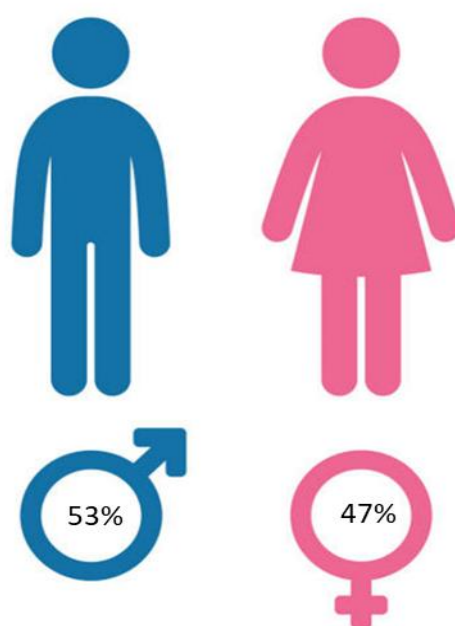


Figure 29: Gender-wise distribution of patients from whom yeast isolates (n=997) were obtained

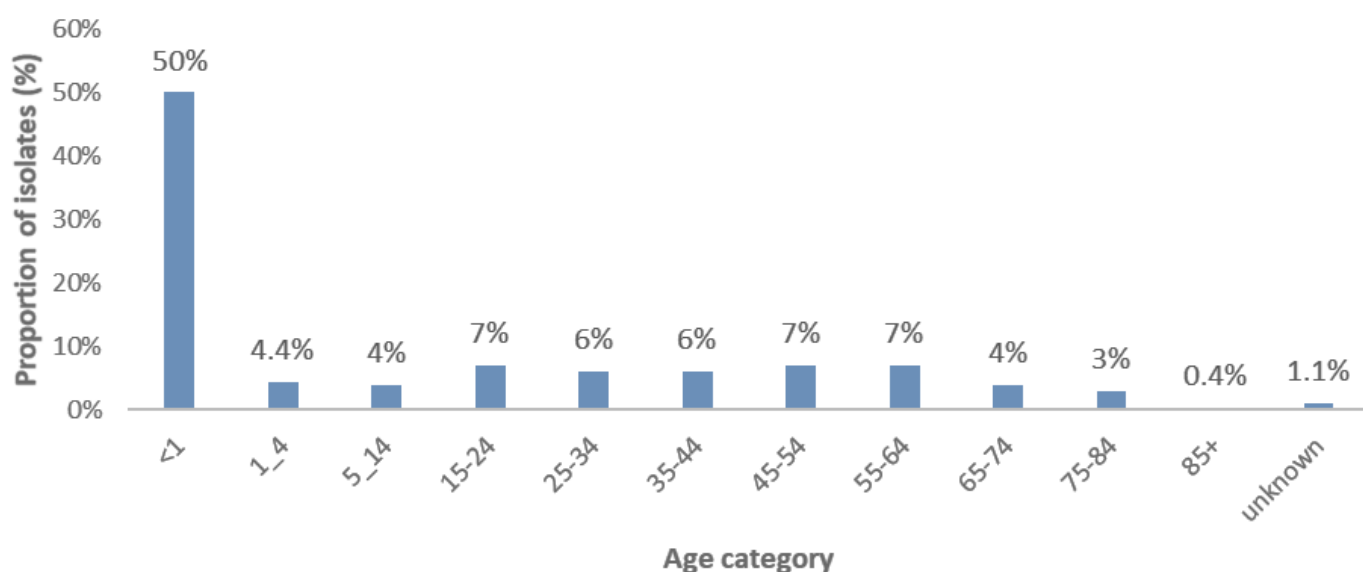


Figure 30: Age distribution of patients from whom yeast isolates were recovered

2.2.3.2 Location wise distribution of isolates

The distribution of fungal isolates across healthcare settings show distinct patterns among different yeast species (Table 16,17,18). Majority of the candidemia cases were from the neonatal intensive care units (NICU) (23%), 5% were from the pediatric intensive care units (PICU) (Table 18). *Candida tropicalis* and *Candida albicans* were the most commonly isolated species. *C. tropicalis* being particularly prevalent in adult intensive care units (ICUs) (53%) and inpatient departments (33%). Majority of isolates from inpatient departments were grown from the blood specimens from pediatric and medicine wards. *Candida albicans* was most commonly isolated from ICUs (47%), indicating its continued clinical significance in critical care. (Table 16,17). Also, *C. parapsilosis* showed a strong presence in ICUs (54%). *C. auris* is a multidrug-resistant species and was primarily found in ICUs (70%) followed by inpatient wards (21%). Overall, ICUs accounted for the highest proportion of isolates for most species, highlighting the critical care unit as a major reservoir for opportunistic fungal infections (Table 16).

Table 16: Distribution of different yeast species by location type in ICU and emergency

Species (n=number of isolates)	Intensive Care Unit (n= 513, 51%)	Emergency (n=110, 11%)
	n (%)	n (%)
<i>Candida tropicalis</i> (n=443)	235 (53%)	51 (12%)
<i>Candida albicans</i> (n=128)	60 (47%)	13 (10%)
<i>Candida parapsilosis</i> (n=94)	50 (54%)	19 (20%)
<i>Pichia kudriavzevii</i> (n=81)	46 (57%)	6
<i>Wickerhamomyces anomalus</i> (n=43)	23 (53%)	4
<i>Candida auris</i> (n=43)	30 (70%)	4
<i>Cyberlindnera jadinii</i> (n=32)	17 (53%)	2
<i>Candida orthopsilosis</i> (n=30)	13 (43%)	4
<i>Candida glabrata</i> (n=22)	12	3
<i>Cyberlindnera fabianii</i> (n=15)	4	
<i>Diutina rugosa</i> (n=16)	6	1
<i>Kodemea ohmeri</i> (n=14)	7	1
<i>Candida viswanathii</i> (n=8)	4	1
<i>Meyerozyma guilliermondii</i> (n=8)	2	
<i>Candida metapsilosis</i> (n=7)	2	1
<i>Clavispora lusitaniae</i> (n=6)	2	1

Table 17: Distribution of different yeast species by location type in Inpatient Wards and Outpatient

Species	Inpatient ward (n= 342,, 34%)								Outpatient (n= 29, 3%)
	PED	MED	NEPH	SUR	OPH	NEU	TRAUMA	OBG/ GYN	
<i>Candida tropicalis</i>	55	24	6	20	2	29	8	1	11
<i>Candida albicans</i>	25	21	1	3		1	2		2
<i>Candida parapsilosis</i>	8	6		2		1		1	7
<i>Pichia kudriavzevii</i>	23	4							2
<i>Wickerhamomyces anomalus</i>	13	2							1
<i>Candida auris</i>	4	3		1			1		-
<i>Cyberlindnera jadinii</i>	12								1
<i>Candida orthopsilosis</i>	5	6		1					1
<i>Candida glabrata</i>	1	3		2				1	-

<i>Cyberlindnera fabianii</i>	7	2		1					1
<i>Diutina rugosa</i>	4	2		1		2			
<i>Kodemea ohmeri</i>	4	2							
<i>Candida viswanathii</i>	2	1							
<i>Meyerozyma guilliermondii</i>	1	2		2					1
<i>Candida metapsilosis</i>	1	1		1					1
<i>Clavispora lusitanae</i>	1	1							
<i>Candida duobushaemulonii</i>	1			2					
<i>Trichosporon asahii</i>		1					1		1
<i>Candida blankii</i>	1								
Total	168	81	7	36	2	33	12	3	29

2.2.3.3 Distribution of isolates across different ICU types

Analysis of yeast isolates distribution across ICU types highlights *C. tropicalis* as the most widespread species, with notable presence in NICU (65%) and PICU (12%). *P. kudriavzevii* and *W. anomalus* were predominantly isolated from NICU with 88% and 68% respectively suggesting strong association with neonatal care. (Table 5).

Table 18: Distribution of different *Candida* species in different types of Intensive Care Units (ICUs)

Type of ICU*	Neonatal and pediatric ICU's			Adult ICU's					
	NICU	PICU	SNCU	MICU	SICU	TICU	GICU	BURN ICU	EICU
<i>Candida tropicalis</i> (n= 185)	121	22	8	12	3	11	3	1	4
<i>Pichia kudriavzevii</i> (n=49)	43	1	3	1	1				
<i>Candida albicans</i> (n=34)	19	2	1	10	1	1			
<i>Candida parapsilosis</i> (n=39)	13	13	3	2	4	1		2	1
<i>Candida auris</i> (n=10)	3	1				6			
<i>Cyberlindnera jadinii</i> (n=11)	3	3	5						

<i>Candida orthopsilosis</i> (n=1)				1					
<i>Wickerhamomyces anomalous</i> (n=22)	15	3	3						1
<i>Candida glabrata</i> (n=9)	2			2	1	2	1	1	
<i>Cyberlindnera fabianii</i> (n=8)	3		4					1	
<i>Diutina rugosa</i> (n=2)	1			1					
<i>Kodemea ohmeri</i> (n=6)	4	1		1					
<i>Candida viswanathii</i> (n=4)	3			1					
<i>Meyerozyma guilliermondii</i> (n=1)						1			
<i>Clavispora lusitania</i> (n=2)	2								
Total	232	46	27	31	10	22	4	5	6

*NICU, Neonatal Intensive Care Unit; PICU, Pediatric Intensive Care Unit; MICU, Medical Intensive Care Unit; SICU, Surgical Intensive Care Unit; TICU, Trauma Intensive Care Unit, SNCU Special Newborn Care Unit, GICU General Intensive Care Unit, BURNICU, Burn Intensive Care Unit, EICU, Emergency Intensive Care Unit

2.2.3.4 Antifungal Susceptibility testing

Candida tropicalis: The azole susceptibility pattern of 443 *C. tropicalis* was assessed, fluconazole exhibited the highest resistance rate at approximately 6% (MIC, ≥ 8 mg/L), followed by voriconazole with a resistance of about 2.3% (MIC, ≥ 1 mg/L). Other antifungals like, itraconazole, posaconazole and amphotericin B, 5-fluorocytosine, micafungin, caspofungin, anidulafungin demonstrated good activity against all *C. tropicalis* isolates.

Candida albicans: The resistance analysis of *C. albicans* (n=128) shows notable resistance to fluconazole (5.5%; MIC, ≥ 8 mg/L). Other azoles, echinocandins, 5-fluorocytosine and amphotericin B exhibited good activity against *C. albicans* isolates.

Candida parapsilosis: The antifungal susceptibility analysis of *C. parapsilosis* (n=94) reveals a pronounced resistance to fluconazole (MIC, ≥ 8 mg/L), with a resistance rate of approximately 19.3%, the highest among all antifungals tested for this species. Whereas, no significant resistance was detected against itraconazole, posaconazole, voriconazole, 5-fluorocytosine, micafungin, caspofungin, anidulafungin.

Pichia kudriavzevii: The antifungal susceptibility pattern of *P. kudriavzevii* (n=81) reveals intrinsic resistance to fluconazole (MIC >32mg/L).

Wickerhamomyces anomalus: As no breakpoints are available for *W. anomalus*, however epidemiological cut-off values (ECV) are given in CLSI M57SEd.4, so we used ECV to categorise the isolates into wild type and non-wild type. All strains (n=43) exhibited low MICs for all the tested antifungals except fluconazole and amphotericin B. Out of 43, 4 isolates (9.3%) were non-wild type as they exhibited MIC value >8 mg/L for fluconazole.

Candida auris: There are currently no established *C. auris*-specific susceptibility breakpoints in CLSI-guidelines. Therefore, we use tentative breakpoints for *C. auris* against different antifungals as recommended by CDC 2024. The antifungal susceptibility analysis of *C. auris* (n=43) showed that 70% of the isolates were resistant to fluconazole (MIC, ≥ 32 mg/L). Notably, a small proportion of isolates (9.7%) shows high MIC value for amphotericin B (MIC, 4 mg/L).

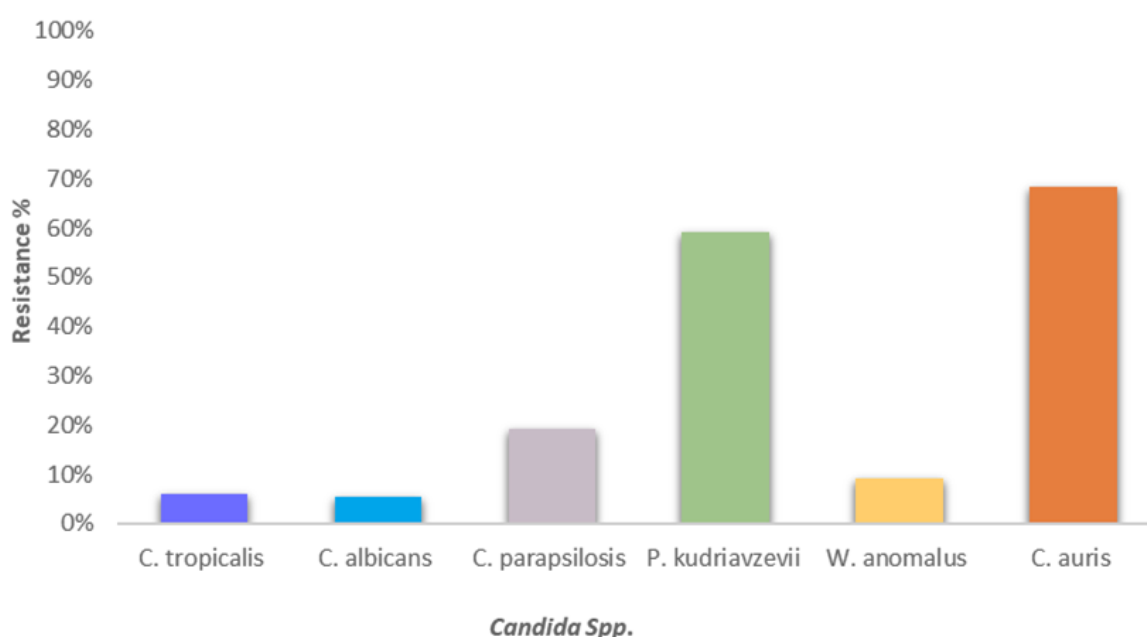


Figure 31: Bar graph representing resistance percentages of *Candida* spp. to fluconazole

2.3 Discussion

A key finding of this surveillance report was the continued predominance of non-*albicans Candida* (NAC) species, which together accounted for the majority of bloodstream isolates. *Candida tropicalis* remained the leading pathogen (45%), followed by *Candida albicans* (13%), *Candida parapsilosis* (10%), *Pichia kudriavzevii* (8%), and *Candida auris* (4%). Age-wise analysis revealed that neonates and infants (<1 year) accounted for nearly 50% of all

bloodstream isolates, emphasizing the disproportionate burden of invasive fungal infections in this vulnerable population. Consistent with this observation, almost 23% of all candidemia cases originated from neonatal intensive care units (NICUs). The recovery of several uncommon yeasts, including *Cyberlindnera jadinii*, *Candida orthopsilosis*, *Clavispora lusitaniae*, *Candida blankii*, *Diutina rugosa*, and *Trichosporon asahii*, further demonstrates the diversity of invasive yeast pathogens and highlights the importance of advanced identification methods such as MALDI-TOF MS and ITS sequencing in routine surveillance. Antifungal susceptibility testing demonstrated that azole resistance, particularly against fluconazole, remains the most important resistance concern. The highest fluconazole resistance was observed in *C. auris* (70%), *Candida parapsilosis* (19%) and *Pichia kudriavzevii* which is intrinsically resistant. Resistance to amphotericin B, echinocandins, and other antifungal agents remained relatively uncommon across most species, although reduced susceptibility was detected in a small proportion of isolates, particularly among *C. auris*. These findings highlight the need for continuous nationwide surveillance, accurate species-level identification, and routine antifungal susceptibility testing to guide effective clinical management and antifungal stewardship.

Annexure I

List of NARS-Net sites that contributed AMR surveillance data for priority bacterial pathogens for the period Jan 2025 to Dec 2025. AMR Surveillance Data for fungal pathogens is from the sites with star (39 sites) in the list below.

1. BJ Medical College, Ahmedabad, Gujarat*
2. BJ Medical college, Pune, Maharashtra
3. Government Medical College and Hospital, Chandigarh*
4. GSVM Medical College, Kanpur, Uttar Pradesh*
5. Lady Hardinge Medical College and Associated hospitals, Delhi*
6. Mysore Medical college, Mysuru, Karnataka*
7. SMS Medical College, Jaipur, Rajasthan*
8. Vardhman Mahavir Medical college and SJ Hospital, Delhi
9. Government Medical College, Thiruvananthapuram, Kerala*
10. KAPV. Government Medical College, Tiruchirappalli, Tamil Nadu*
11. Gauhati Medical College and Hospital, Guwahati, Assam*
12. NEIGRIHMS, Shillong, Meghalaya*
13. MGM College and Hospital, Indore, Madhya Pradesh
14. Indira Gandhi Medical College, Shimla, Himachal Pradesh*
15. Govt. Medical College and Hospital, Aurangabad, Maharashtra*
16. Osmania Medical College, Hyderabad, Telangana
17. Guntur Medical College, Guntur, Andhra Pradesh
18. Agartala Govt. Medical College, Agartala, Tripura
19. SCB Medical College & Hospital, Cuttack, Odisha
20. Government Medical College & Hospital, Jammu, Jammu and Kashmir*
21. Pandit Bhagwat Dayal Sharma, Post Graduate Institute of Medical Sciences (PGIMS)
Rohtak, Haryana*
22. Rajendra Institute of Medical Sciences, Ranchi, Jharkhand*
23. Indira Gandhi Institute of Medical Sciences, Sheikpura, Patna, Bihar*
24. Government Medical College, Haldwani, Uttarakhand*
25. Pt. Jawaharlal Nehru Memorial Medical College, Raipur, Chhattisgarh*
26. Gandhi Medical College, Bhopal, Madhya Pradesh*
27. Calcutta School of Tropical Medicine, Kolkata, West Bengal*
28. GMERS Medical College and Civil Hospital, Valsad, Gujarat*

29. Lala Lajpat Rai Memorial (LLRM) Medical College, Meerut, Uttar Pradesh *
30. Coimbatore Medical College & Hospital, Coimbatore, Tamil Nadu
31. Maulana Azad Medical College (MAMC) and Associated Hospitals, Delhi *
32. Sardar Patel Medical College (SPMC) and Hospital, Bikaner, Rajasthan *
33. Karnataka Institute of Medical Sciences (KIMS), Hubli, Karnataka *
34. Indira Gandhi Medical College & Research Institute (IGMC & RI) Puducherry *
35. NAMO Medical Education and Research Institute (MERI), Silvassa, Dadra & Nagar Haveli *
36. Goa Medical College & Hospital, Bambolim, Goa
37. STNM Medical College & Hospital, Gangtok, Sikkim *
38. Government Medical College, Patiala, Punjab
39. Zoram Medical College, Falkawn, Mizoram
40. Andaman & Nicobar Islands Institute of Medical Sciences (ANIIMS), Andaman & Nicobar Islands
41. Rabindranath Tagore Medical College, Udaipur, Rajasthan *
42. Toma Riba Institute of Health and Medical Sciences, Naharlugan, Arunachal Pradesh
43. Jawahar Lal Nehru Institute of Medical Sciences (JNIMS), Manipur
44. Govt. Medical College Srinagar Jammu and Kashmir
45. Andhra Medical College Vishakhapatnam, Andhra Pradesh *
46. Vijayanagar Institute of Medical Sciences Ballari, Karnataka *
47. Moti Lal Nehru Medical College Allahabad, Uttar Pradesh
48. Burdwan Medical College & Hospital Burdwan, West Bengal *
49. Grant Govt Medical College & Sir JJ Group of Hospitals, Byculla, Mumbai *
50. Pt. Raghunath Murmu Medical College & Hospital Baripada, Odisha
51. Government Medical College, Thrissur, Kerala *
52. Bangalore Medical College and Research Institute, Bengaluru, Karnataka *
53. Kakatiya Medical College, Warangal, Telangana
54. Madras Medical College, Chennai, Tamil Nadu *
55. S.V medical College, Tirupati, Andhra Pradesh
56. Gajra Raja Medical College, Gwalior, Madhya Pradesh *
57. Jorhat Medical College and Hospital, Jorhat, Assam *
58. University College of Medical Sciences & GTB Hospital, Delhi *
59. Pandit Dindyal Upadhyay Medical College, Rajkot, Gujarat
60. Netaji Subash Chandra Bose Medical College, Jabalpur, Madhya Pradesh
61. Govt Medical College, Nagpur, Maharashtra
62. Government Medical College, Kota, Rajasthan